

Electric Circuit Materials

Question-Wise Past Paper Solutions

Bachelor of Engineering (BE Level)

Arranged Question-by-Question Across 4 Exam Papers:

1. 2083 Baishakh — Back Exam
2. 2082 Bhadra — Regular Exam
3. 2082 Baishakh — Back Exam
4. 2081 Ashwin — Regular Exam

Question 1 $\rightarrow$ Question 2 $\rightarrow \dots \rightarrow$ Question 10

Contents

1	Question 1 (Nodal Analysis, Mesh Analysis & Matrix Methods)	3
1.1	2083 Baishakh Exam Paper	3
1.2	2082 Bhadra Exam Paper	5
1.3	2082 Baishakh Exam Paper	7
1.4	2081 Ashwin Exam Paper	9
2	Question 2 (Initial Conditions & Transient Response in RLC Circuits)	11
2.1	2083 Baishakh Exam Paper	11
2.2	2082 Bhadra Exam Paper	13
2.3	2082 Baishakh Exam Paper	15
2.4	2081 Ashwin Exam Paper	17
3	Question 3 (Classical Transient Analysis of RC, RL & RLC Circuits)	18
3.1	2083 Baishakh Exam Paper	18
3.2	2082 Bhadra Exam Paper	20
3.3	2082 Baishakh Exam Paper	22
3.4	2081 Ashwin Exam Paper	24
4	Question 4 (Laplace Transform Applications to Circuit Analysis)	26
4.1	2083 Baishakh Exam Paper	26
4.2	2082 Bhadra Exam Paper	27
4.3	2082 Baishakh Exam Paper	28
4.4	2081 Ashwin Exam Paper	30
5	Question 5 (Two-Port Networks & Bode Plots)	32
5.1	2083 Baishakh Exam Paper	32
5.2	2082 Bhadra Exam Paper	34
5.3	2082 Baishakh Exam Paper	36
5.4	2081 Ashwin Exam Paper	38
6	Question 6 (Two-Port Parameters & Frequency Response / Bode Plots)	39
6.1	2083 Baishakh Exam Paper	39
6.2	2082 Bhadra Exam Paper	41
6.3	2082 Baishakh Exam Paper	43
6.4	2081 Ashwin Exam Paper	45
7	Question 7 (Transformers & DC Machines)	47
7.1	2083 Baishakh Exam Paper	47
7.2	2082 Bhadra Exam Paper	49
7.3	2082 Baishakh Exam Paper	50
7.4	2081 Ashwin Exam Paper	52
8	Question 8 (DC Generators, DC Motors & Magnetic Circuits)	53
8.1	2083 Baishakh Exam Paper	53
8.2	2082 Bhadra Exam Paper	55
8.3	2082 Baishakh Exam Paper	56
8.4	2081 Ashwin Exam Paper	57

9	Question 9 (Induction Motors & Transformer Efficiency)	58
9.1	2083 Baishakh Exam Paper	58
9.2	2082 Bhadra Exam Paper	59
9.3	2082 Baishakh Exam Paper	60
9.4	2081 Ashwin Exam Paper	61
10	Question 10 (Short Notes & Conceptual Explanations)	63
10.1	2083 Baishakh Exam Paper	63
10.2	2082 Bhadra Exam Paper	65
10.3	2082 Baishakh Exam Paper	67
10.4	2081 Ashwin Exam Paper	68

1 Question 1 (Nodal Analysis, Mesh Analysis & Matrix Methods)

1.1 2083 Baishakh Exam Paper

Question 1

Find the current in 10 V source using nodal analysis.

Solution:

1. Circuit Setup & Node Identification:

- **Reference Node (Ground):** Bottom wire taken as 0 V.
- **Essential Nodes:**
 - v_1 : Node above the 2Ω resistor.
 - v_2 : Node between the 10 V source, 6Ω resistor, 4Ω resistor, and the dependent source.
 - v_3 : Node above the 3Ω resistor.
- **Dependent Source:** Voltage source $5i$ between node v_2 and node v_3 .
- **Controlling Current i :** Flows downward through the 2Ω resistor:

$$i = \frac{v_1}{2}$$

- **Supernode 1 (between v_1 and v_2):** Connected via a 10 V independent source:

$$v_2 - v_1 = 10 \implies v_2 = v_1 + 10$$

- **Supernode 2 (between v_2 and v_3):** Connected via a dependent voltage source $5i$:

$$v_3 - v_2 = 5i = 5 \left(\frac{v_1}{2} \right) = 2.5v_1$$

Substituting $v_2 = v_1 + 10$:

$$v_3 = (v_1 + 10) + 2.5v_1 = 3.5v_1 + 10$$

2. Formulating the Nodal Equation: Since nodes v_1 , v_2 , and v_3 are connected together via voltage sources, they form a single combined **Supernode** (v_1, v_2, v_3).

Applying Kirchhoff's Current Law (KCL) at the Supernode:

$$\frac{v_1}{2} + \frac{v_2 - v_3}{6} + \frac{v_2}{4} + \frac{v_3 - v_2}{6} + \frac{v_3}{3} = 0$$

Simplifying (the 6Ω resistor terms cancel out internally across v_2 and v_3):

$$\frac{v_1}{2} + \frac{v_2}{4} + \frac{v_3}{3} = 0$$

Substitute v_2 and v_3 in terms of v_1 :

$$\frac{v_1}{2} + \frac{v_1 + 10}{4} + \frac{3.5v_1 + 10}{3} = 0$$

Multiply the entire equation by 12 to clear denominators:

$$6v_1 + 3(v_1 + 10) + 4(3.5v_1 + 10) = 0$$

$$6v_1 + 3v_1 + 30 + 14v_1 + 40 = 0$$

$$23v_1 + 70 = 0 \implies v_1 = -\frac{70}{23} \text{ V} \approx -3.043 \text{ V}$$

3. Calculating Node Voltages:

- $v_1 = -3.043 \text{ V}$
- $v_2 = v_1 + 10 = -3.043 + 10 = 6.957 \text{ V}$
- $v_3 = v_2 + 2.5v_1 = 6.957 + 2.5(-3.043) = -0.6505 \text{ V}$

4. Finding Current through the 10 V Source: Applying KCL specifically at Node v_1 :

$$\frac{v_1}{2} + \frac{v_1 - v_3}{6} + I_{10\text{V}} = 0$$

$$I_{10\text{V}} = -\left(\frac{v_1}{2} + \frac{v_1 - v_3}{6}\right)$$

Substituting values:

$$I_{10\text{V}} = -\left(\frac{-3.043}{2} + \frac{-3.043 - (-0.6505)}{6}\right)$$

$$I_{10\text{V}} = -(-1.5215 - 0.39875) = \mathbf{1.92 \text{ A}}$$

Final Answer: The current flowing through the 10 V source (from node v_1 to node v_2) is **1.92 A**.

1.2 2082 Bhadra Exam Paper

Question 1

Find the current I_x and power delivered by the dependent voltage source in the circuit shown using Mesh Analysis.

Solution:

1. Circuit Identification & Mesh Setup:

- **Loop 1 (Top Left):** Contains 100 V source and $5\ \Omega$ resistor.
- **Loop 2 (Bottom Left):** Contains $10I_x$ dependent source and $10\ \Omega$ resistor.
- **Loop 3 (Middle):** Contains $15\ \Omega$ resistor and $50\ \Omega$ resistor (I_x branch).
- **Loop 4 (Right):** Contains $25\ \Omega$ resistor.

From the circuit diagram:

- Controlling current I_x is the downward current through the $50\ \Omega$ resistor branch.
- $I_x = I_3 - I_2$ (where I_3 is bottom-right mesh current and I_2 is top-right mesh current).

Assigning standard mesh currents I_1, I_2, I_3 :

1. **Mesh 1 (Left loop):** I_1 clockwise.
2. **Mesh 2 (Top-Right loop through $5\ \Omega, 15\ \Omega, 25\ \Omega$):** I_2 clockwise.
3. **Mesh 3 (Bottom-Right loop through $10\ \Omega, 50\ \Omega$):** I_3 clockwise.

2. Formulating Mesh Equations:

Mesh 1 Equation:

$$100 + 10I_x + 10(I_1 - I_3) + 5(I_1 - I_2) = 0$$

$$15I_1 - 5I_2 - 10I_3 + 10I_x = -100$$

Substitute $I_x = I_3 - I_2$:

$$15I_1 - 5I_2 - 10I_3 + 10(I_3 - I_2) = -100$$

$$15I_1 - 15I_2 = -100 \implies 3I_1 - 3I_2 = -20 \quad \text{--- (Eq. 1)}$$

Mesh 2 Equation:

$$5(I_2 - I_1) + 15(I_2 - I_3) + 25I_2 = 0$$

$$-5I_1 + 45I_2 - 15I_3 = 0 \implies -I_1 + 9I_2 - 3I_3 = 0 \quad \text{--- (Eq. 2)}$$

Mesh 3 Equation:

$$\begin{aligned}
10(I_3 - I_1) + 15(I_3 - I_2) + 50I_3 - 10I_x &= 0 \\
-10I_1 - 15I_2 + 75I_3 - 10(I_3 - I_2) &= 0 \\
-10I_1 - 5I_2 + 65I_3 &= 0 \implies -2I_1 - I_2 + 13I_3 = 0 \quad \text{--- (Eq. 3)}
\end{aligned}$$

3. Solving the System of Equations: From **Eq. 1**:

$$I_1 = I_2 - \frac{20}{3}$$

Substitute I_1 into **Eq. 2**:

$$\begin{aligned}
-\left(I_2 - \frac{20}{3}\right) + 9I_2 - 3I_3 &= 0 \implies 8I_2 - 3I_3 = -\frac{20}{3} \\
24I_2 - 9I_3 &= -20 \quad \text{--- (Eq. 4)}
\end{aligned}$$

Substitute I_1 into **Eq. 3**:

$$\begin{aligned}
-2\left(I_2 - \frac{20}{3}\right) - I_2 + 13I_3 &= 0 \implies -3I_2 + 13I_3 = -\frac{40}{3} \\
-9I_2 + 39I_3 &= -40 \quad \text{--- (Eq. 5)}
\end{aligned}$$

Solving **Eq. 4** and **Eq. 5**: Multiply **Eq. 4** by 39 and **Eq. 5** by 9:

$$\begin{aligned}
936I_2 - 351I_3 &= -780 \\
-81I_2 + 351I_3 &= -360
\end{aligned}$$

Adding both equations:

$$855I_2 = -1140 \implies I_2 = -1.333 \text{ A} = -\frac{4}{3} \text{ A}$$

Now finding I_3 :

$$24\left(-\frac{4}{3}\right) - 9I_3 = -20 \implies -32 - 9I_3 = -20 \implies 9I_3 = -12 \implies I_3 = -\frac{4}{3} \text{ A}$$

Now finding I_1 :

$$I_1 = -\frac{4}{3} - \frac{20}{3} = -8 \text{ A}$$

4. Calculated Parameters:**1. Current I_x :**

$$I_x = I_3 - I_2 = -\frac{4}{3} - \left(-\frac{4}{3}\right) = 0 \text{ A}$$

2. Power Delivered by Dependent Source ($10I_x$):

- Since $I_x = 0 \text{ A}$, dependent voltage $V_{\text{dep}} = 10I_x = 0 \text{ V}$.
- Power $P = V_{\text{dep}} \times I_{\text{branch}} = 0 \times I_{\text{branch}} = 0 \text{ W}$.

Final Answer: $I_x = 0 \text{ A}$ and Power delivered by dependent source = 0 W .

1.3 2082 Baishakh Exam Paper

Question 1

In the given circuit, determine V_x and power consumed by it using mesh analysis and solve by matrix method.

Solution:

1. Circuit Description & Mesh Setup:

- Independent current source: -10 A (equivalent to 10 A upwards).
- Dependent current source: $\frac{V_x}{10}$.
- Resistors: $1\ \Omega$, $2\ \Omega$, $3\ \Omega$, $5\ \Omega$.
- Voltage V_x across the $3\ \Omega$ resistor:

$$V_x = 3(I_3 - I_2)$$

2. Mesh Definition & Equations: Let mesh currents be I_1, I_2, I_3 clockwise for Mesh 1 (left), Mesh 2 (top-right), and Mesh 3 (bottom-right):

1. Mesh 1 Constraint:

$$I_1 = -10\text{ A}$$

2. Dependent Source Constraint (between Mesh 1 and Mesh 3):

$$I_1 - I_3 = \frac{V_x}{10} = \frac{3(I_3 - I_2)}{10}$$

Substituting $I_1 = -10$:

$$-10 - I_3 = \frac{3(I_3 - I_2)}{10} \implies -100 - 10I_3 = 3I_3 - 3I_2$$

$$3I_2 - 13I_3 = 100 \quad \text{--- (Eq. 1)}$$

3. Mesh 2 Equation (Top-right loop through $1\ \Omega, 2\ \Omega, 3\ \Omega$):

$$1(I_2 - I_1) + 2I_2 + 3(I_2 - I_3) = 0$$

$$1(I_2 - (-10)) + 2I_2 + 3I_2 - 3I_3 = 0$$

$$6I_2 - 3I_3 = -10 \quad \text{--- (Eq. 2)}$$

3. Matrix System Formulation: Formulating the matrix equation $\mathbf{A} \cdot \mathbf{I} = \mathbf{B}$ for variables $\begin{bmatrix} I_2 \\ I_3 \end{bmatrix}$:

$$\begin{bmatrix} 6 & -3 \\ 3 & -13 \end{bmatrix} \begin{bmatrix} I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} -10 \\ 100 \end{bmatrix}$$

4. Solving Matrix using Cramer's Rule:

$$\Delta = \begin{vmatrix} 6 & -3 \\ 3 & -13 \end{vmatrix} = (6)(-13) - (-3)(3) = -78 + 9 = -69$$

$$\Delta_2 = \begin{vmatrix} -10 & -3 \\ 100 & -13 \end{vmatrix} = (-10)(-13) - (-3)(100) = 130 + 300 = 430$$

$$\Delta_3 = \begin{vmatrix} 6 & -10 \\ 3 & 100 \end{vmatrix} = (6)(100) - (-10)(3) = 600 + 30 = 630$$

Solving mesh currents:

$$I_2 = \frac{\Delta_2}{\Delta} = \frac{430}{-69} \approx -6.232 \text{ A}$$

$$I_3 = \frac{\Delta_3}{\Delta} = \frac{630}{-69} \approx -9.130 \text{ A}$$

5. Final Calculations:**1. Voltage V_x :**

$$V_x = 3(I_3 - I_2) = 3(-9.130 - (-6.232)) = 3(-2.898) = -8.694 \text{ V}$$

2. Power Consumed by V_x (Power across 3Ω Resistor):

$$P_{3\Omega} = \frac{V_x^2}{R} = \frac{(-8.694)^2}{3} = \frac{75.585}{3} = 25.20 \text{ W}$$

Final Answer: $V_x = -8.69 \text{ V}$ and Power consumed = **25.20 W**.

1.4 2081 Ashwin Exam Paper

Question 1

Using Nodal analysis and matrix solution method, find V_x and the current in 2Ω resistance.

Solution:

1. Circuit Setup & Node Definitions:

- **Node 1 (V_1):** Above 3Ω resistor.
- **Node 2 (V_2):** Right of 4Ω resistor, above 1Ω resistor.
- **Reference (Ground):** Bottom wire.
- **Dependent Source:** $4V_x$ current source entering Node V_2 .
- **Branch Voltage V_x :** Voltage across 4Ω resistor:

$$V_x = (V_1 - 5) - V_2$$

2. Nodal Equations:

Node 1 (V_1) KCL:

$$\frac{V_1}{3} + 5 + \frac{V_1 - V_2}{2} + \frac{(V_1 - 5) - V_2}{4} = 0$$

Multiply by 12:

$$4V_1 + 60 + 6(V_1 - V_2) + 3(V_1 - 5 - V_2) = 0$$

$$4V_1 + 60 + 6V_1 - 6V_2 + 3V_1 - 15 - 3V_2 = 0$$

$$13V_1 - 9V_2 = -45 \quad \text{--- (Eq. 1)}$$

Node 2 (V_2) KCL:

$$\frac{V_2}{1} + \frac{V_2 - V_1}{2} + \frac{V_2 - (V_1 - 5)}{4} - 4V_x = 0$$

Substitute $V_x = V_1 - 5 - V_2$:

$$V_2 + \frac{V_2 - V_1}{2} + \frac{V_2 - V_1 + 5}{4} - 4(V_1 - 5 - V_2) = 0$$

Multiply by 4:

$$4V_2 + 2(V_2 - V_1) + (V_2 - V_1 + 5) - 16(V_1 - 5 - V_2) = 0$$

$$4V_2 + 2V_2 - 2V_1 + V_2 - V_1 + 5 - 16V_1 + 80 + 16V_2 = 0$$

$$-19V_1 + 23V_2 = -85 \quad \text{--- (Eq. 2)}$$

3. Matrix Formulation & Solution: Writing the matrix system $\mathbf{A} \cdot \mathbf{V} = \mathbf{B}$:

$$\begin{bmatrix} 13 & -9 \\ -19 & 23 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -45 \\ -85 \end{bmatrix}$$

Determinant Calculation:

$$\Delta = \begin{vmatrix} 13 & -9 \\ -19 & 23 \end{vmatrix} = (13)(23) - (-9)(-19) = 299 - 171 = 128$$

$$\Delta_1 = \begin{vmatrix} -45 & -9 \\ -85 & 23 \end{vmatrix} = (-45)(23) - (-9)(-85) = -1035 - 765 = -1800$$

$$\Delta_2 = \begin{vmatrix} 13 & -45 \\ -19 & -85 \end{vmatrix} = (13)(-85) - (-45)(-19) = -1105 - 855 = -1960$$

Node Voltages:

$$V_1 = \frac{\Delta_1}{\Delta} = \frac{-1800}{128} = -14.0625 \text{ V}$$

$$V_2 = \frac{\Delta_2}{\Delta} = \frac{-1960}{128} = -15.3125 \text{ V}$$

4. Final Calculations:

1. **Voltage V_x :**

$$V_x = V_1 - 5 - V_2 = -14.0625 - 5 - (-15.3125) = -\mathbf{3.75 \text{ V}}$$

2. **Current through 2Ω Resistor ($I_{2\Omega}$):**

$$I_{2\Omega} = \frac{V_1 - V_2}{2} = \frac{-14.0625 - (-15.3125)}{2} = \frac{1.25}{2} = \mathbf{0.625 \text{ A}}$$

Final Answer: $V_x = -\mathbf{3.75 \text{ V}}$ and Current in 2Ω resistor = $\mathbf{0.625 \text{ A}}$.

2 Question 2 (Initial Conditions & Transient Response in RLC Circuits)

2.1 2083 Baishakh Exam Paper

Question 2

Obtain the value of $i_1, i_2, \frac{di_1}{dt}, \frac{di_2}{dt}$ at $t = 0^+$, if the switch is closed at $t = 0$ in the circuit shown.

Solution:

1. Initial Conditions at $t = 0^-$: Before $t = 0$, the switch is open and the circuit is in steady state with zero initial energy stored:

- $i_1(0^-) = 0 \text{ A}$
- $i_2(0^-) = 0 \text{ A}$
- Initial inductor current: $i_L(0^-) = i_1(0^-) = 0 \text{ A}$
- Initial capacitor voltage: $v_C(0^-) = 0 \text{ V}$

Since current through an inductor and voltage across a capacitor cannot change instantaneously:

$$i_1(0^+) = i_L(0^+) = i_L(0^-) = 0 \text{ A}$$

$$v_C(0^+) = v_C(0^-) = 0 \text{ V}$$

To find $i_2(0^+)$, consider the second loop containing the 10Ω resistor and the capacitor:

- Voltage across the parallel branch is $v_C(0^+) = 0 \text{ V}$.
- Therefore, $i_2(0^+) = \frac{v_C(0^+)}{10} = 0 \text{ A}$.

2. Circuit Response at $t = 0^+$: At $t = 0^+$, the switch is closed. Component behavior:

- Inductor acts as an **open circuit** ($i_1(0^+) = 0$).
- Capacitor acts as a **short circuit** ($v_C(0^+) = 0$).

Applying KVL to Loop 1 at $t = 0^+$:

$$10 - L \left. \frac{di_1}{dt} \right|_{t=0^+} - 10(i_1(0^+) - i_2(0^+)) = 0$$

Substitute $L = 1 \text{ H}$, $i_1(0^+) = 0 \text{ A}$, $i_2(0^+) = 0 \text{ A}$:

$$10 - (1) \left. \frac{di_1}{dt} \right|_{t=0^+} - 10(0 - 0) = 0 \implies \left. \frac{di_1}{dt} \right|_{t=0^+} = 10 \text{ A/s}$$

Applying KVL to Loop 2 at $t = 0^+$:

$$10(i_2(t) - i_1(t)) + 10i_2(t) + v_C(t) = 0$$

Differentiating with respect to t :

$$10 \left(\frac{di_2}{dt} - \frac{di_1}{dt} \right) + 10 \frac{di_2}{dt} + \frac{i_2(t)}{C} = 0$$

At $t = 0^+$, substitute $i_2(0^+) = 0 \text{ A}$ and $\left. \frac{di_1}{dt} \right|_{t=0^+} = 10 \text{ A/s}$:

$$10 \left(\left. \frac{di_2}{dt} \right|_{t=0^+} - 10 \right) + 10 \left. \frac{di_2}{dt} \right|_{t=0^+} + 0 = 0$$

$$20 \left. \frac{di_2}{dt} \right|_{t=0^+} - 100 = 0 \implies \left. \frac{di_2}{dt} \right|_{t=0^+} = 5 \text{ A/s}$$

Final Answer: $i_1(0^+) = 0 \text{ A}$, $i_2(0^+) = 0 \text{ A}$, $\left. \frac{di_1}{dt} \right|_{t=0^+} = 10 \text{ A/s}$, $\left. \frac{di_2}{dt} \right|_{t=0^+} = 5 \text{ A/s}$.

2.2 2082 Bhadra Exam Paper

Question 2

Explain the behavior of inductor when subjected to switching. In the network, the switch is opened at $t = 0$. Calculate v , $\frac{dv}{dt}$ and $\frac{d^2v}{dt^2}$ at $t = 0^+$ if $I = 2 \text{ A}$, $R = 200 \Omega$ and $L = 1 \text{ H}$. Given that v is the voltage across inductor.

Solution:

Part A: Behavior of Inductor Subjected to Switching An inductor stores energy in its magnetic field ($E = \frac{1}{2}LI^2$). The current through an inductor cannot change instantaneously because an abrupt jump in current ($\frac{di}{dt} \rightarrow \infty$) would induce an infinite voltage ($v = L\frac{di}{dt}$), violating physical energy conservation. Hence:

$$i_L(0^+) = i_L(0^-)$$

Part B: Mathematical Calculations

1. Initial Conditions ($t = 0^-$): Before $t = 0$, switch K is closed, short-circuiting the parallel $R - L$ branch:

- Current through inductor: $i_L(0^-) = 0 \text{ A}$
- Continuity condition: $i_L(0^+) = i_L(0^-) = 0 \text{ A}$

2. Finding $v(0^+)$: At $t = 0^+$, switch K is opened. The constant current source $I = 2 \text{ A}$ feeds into the parallel combination of R and L :

$$I = i_R(t) + i_L(t)$$

At $t = 0^+$:

$$i_R(0^+) = I - i_L(0^+) = 2 - 0 = 2 \text{ A}$$

The voltage across the inductor $v(t)$ equals the voltage across the resistor R :

$$v(0^+) = R \cdot i_R(0^+) = 200 \times 2 = \mathbf{400 \text{ V}}$$

3. Finding $\frac{dv}{dt}\big|_{t=0^+}$: We know $v(t) = L\frac{di_L}{dt}$ and $i_R(t) = I - i_L(t) \implies v(t) = R(I - i_L(t))$.

Differentiating $v(t) = R(I - i_L(t))$ with respect to t :

$$\frac{dv}{dt} = -R\frac{di_L}{dt}$$

Since $v(t) = L\frac{di_L}{dt} \implies \frac{di_L}{dt} = \frac{v(t)}{L}$:

$$\frac{dv}{dt} = -R\left(\frac{v(t)}{L}\right) = -\frac{R}{L}v(t)$$

At $t = 0^+$:

$$\frac{dv}{dt}\bigg|_{t=0^+} = -\frac{200}{1} \times 400 = \mathbf{-80,000 \text{ V/s}} \quad (\text{or } -8 \times 10^4 \text{ V/s})$$

4. Finding $\left. \frac{d^2v}{dt^2} \right|_{t=0^+}$: Differentiating $\frac{dv}{dt} = -\frac{R}{L}v(t)$ once more:

$$\frac{d^2v}{dt^2} = -\frac{R}{L} \frac{dv}{dt}$$

At $t = 0^+$:

$$\left. \frac{d^2v}{dt^2} \right|_{t=0^+} = -\frac{200}{1} \times (-80,000) = \mathbf{16,000,000 \text{ V/s}^2} \quad (\text{or } 1.6 \times 10^7 \text{ V/s}^2)$$

Final Answer: $v(0^+) = \mathbf{400 \text{ V}}$, $\left. \frac{dv}{dt} \right|_{t=0^+} = \mathbf{-80,000 \text{ V/s}}$, $\left. \frac{d^2v}{dt^2} \right|_{t=0^+} = \mathbf{1.6 \times 10^7 \text{ V/s}^2}$.

2.3 2082 Baishakh Exam Paper

Question 2

In the network shown below, the switch has been in position a for a long time. At $t = 0$, it is suddenly thrown off to position b . Find current through each element, $\frac{di_L}{dt}$ of coil and $\frac{dv_C}{dt}$ of capacitor at $t = 0^+$.

Solution:

1. Steady-State Analysis at $t = 0^-$ (Switch at Position a):

- Switch at position a for $t < 0$.
- Inductor acts as a **short circuit** and Capacitor acts as an **open circuit**.
- Voltage source connected = 12 V.

Calculating initial values:

1. **Inductor Branch:** Connected through 2Ω and 4Ω resistors in series with 12 V:

$$i_L(0^-) = \frac{12}{2+4} = \frac{12}{6} = 2 \text{ A}$$

2. **Capacitor Branch:** Open circuit. Voltage across capacitor v_C is the voltage across the 4Ω resistor:

$$v_C(0^-) = i_L(0^-) \times 4 = 2 \times 4 = 8 \text{ V}$$

By continuity conditions at $t = 0^+$:

$$i_L(0^+) = i_L(0^-) = 2 \text{ A}, \quad v_C(0^+) = v_C(0^-) = 8 \text{ V}$$

2. **Circuit Response at $t = 0^+$ (Switch at Position b):** When switch moves to position b , the 12 V supply and 2Ω resistor are disconnected. The remaining active network consists of the 6Ω resistor, 2 F capacitor, 4Ω resistor, and 3 H inductor.

Current through each element at $t = 0^+$:

- **Inductor current:** $i_L(0^+) = 2 \text{ A}$
- **Capacitor voltage:** $v_C(0^+) = 8 \text{ V}$
- **Current through 6Ω resistor ($i_{6\Omega}$):** Connected directly across the 2 F capacitor branch at position b :

$$i_{6\Omega}(0^+) = \frac{v_C(0^+)}{6} = \frac{8}{6} = \mathbf{1.333 \text{ A}}$$

- **Current through 4Ω resistor ($i_{4\Omega}$):** In series with inductor branch:

$$i_{4\Omega}(0^+) = i_L(0^+) = \mathbf{2 \text{ A}}$$

- **Capacitor current (i_C):** Applying KCL at the node above the capacitor:

$$i_C(0^+) + i_{6\Omega}(0^+) + i_L(0^+) = 0$$

$$i_C(0^+) = -i_{6\Omega}(0^+) - i_L(0^+) = -1.333 - 2 = \mathbf{-3.333 \text{ A}}$$

Derivatives at $t = 0^+$:

1. **For Inductor ($\frac{di_L}{dt}$):** Using KVL around the loop containing the capacitor, 4Ω resistor, and 3 H inductor:

$$v_C(t) - 4i_L(t) - L\frac{di_L}{dt} = 0$$

$$8 - 4(2) - 3\left.\frac{di_L}{dt}\right|_{t=0^+} = 0 \implies \left.\frac{di_L}{dt}\right|_{t=0^+} = \mathbf{0\text{ A/s}}$$

2. **For Capacitor ($\frac{dv_C}{dt}$):** Using the capacitor relation $i_C = C\frac{dv_C}{dt}$:

$$\left.\frac{dv_C}{dt}\right|_{t=0^+} = \frac{i_C(0^+)}{C} = \frac{-3.333}{2} = \mathbf{-1.667\text{ V/s}}$$

Final Answer: $i_L(0^+) = \mathbf{2\text{ A}}$, $i_{4\Omega}(0^+) = \mathbf{2\text{ A}}$, $i_{6\Omega}(0^+) = \mathbf{1.333\text{ A}}$, $i_C(0^+) = \mathbf{-3.333\text{ A}}$,
 $\left.\frac{di_L}{dt}\right|_{t=0^+} = \mathbf{0\text{ A/s}}$, $\left.\frac{dv_C}{dt}\right|_{t=0^+} = \mathbf{-1.667\text{ V/s}}$.

2.4 2081 Ashwin Exam Paper

Question 2

Obtain the value of $i_1, i_2, \frac{di_1}{dt}, \frac{di_2}{dt}$ at $t = 0^+$, if the switch is closed at $t = 0$ in the circuit shown below.

Solution:

1. Initial Conditions ($t = 0^-$): Before $t = 0$, the switch is open and the circuit is completely unenergized:

- Initial inductor current: $i_1(0^-) = 0 \text{ A}$
- Initial capacitor voltage: $v_C(0^-) = 0 \text{ V}$

By continuity conditions:

$$i_1(0^+) = i_1(0^-) = 0 \text{ A}$$

$$v_C(0^+) = v_C(0^-) = 0 \text{ V}$$

Since the capacitor has 0 V across it at $t = 0^+$:

$$i_2(0^+) = \frac{v_C(0^+)}{10 \Omega} = 0 \text{ A}$$

2. Circuit Response at $t = 0^+$: When the switch closes at $t = 0$, the 10 V DC source is connected to the circuit:

- Inductor $L = 1 \text{ H}$ acts as an **open circuit** ($i_1 = 0$).
- Capacitor $C = 1 \text{ F}$ acts as a **short circuit** ($v_C = 0$).

Applying KVL in Loop 1 at $t = 0^+$:

$$10 - L \left. \frac{di_1}{dt} \right|_{t=0^+} - 10(i_1(0^+) - i_2(0^+)) = 0$$

Substitute $L = 1 \text{ H}$, $i_1(0^+) = 0 \text{ A}$, $i_2(0^+) = 0 \text{ A}$:

$$10 - (1) \left. \frac{di_1}{dt} \right|_{t=0^+} - 0 = 0 \implies \left. \frac{di_1}{dt} \right|_{t=0^+} = 10 \text{ A/s}$$

Applying KVL in Loop 2 at $t = 0^+$:

$$10(i_2(t) - i_1(t)) + 10i_2(t) + v_C(t) = 0 \implies 20i_2(t) - 10i_1(t) + v_C(t) = 0$$

Differentiating with respect to t :

$$20 \frac{di_2}{dt} - 10 \frac{di_1}{dt} + \frac{i_2(t)}{C} = 0$$

At $t = 0^+$, substitute $\left. \frac{di_1}{dt} \right|_{t=0^+} = 10 \text{ A/s}$ and $i_2(0^+) = 0 \text{ A}$:

$$20 \left. \frac{di_2}{dt} \right|_{t=0^+} - 10(10) + 0 = 0 \implies 20 \left. \frac{di_2}{dt} \right|_{t=0^+} = 100 \implies \left. \frac{di_2}{dt} \right|_{t=0^+} = 5 \text{ A/s}$$

Final Answer: $i_1(0^+) = 0 \text{ A}$, $i_2(0^+) = 0 \text{ A}$, $\left. \frac{di_1}{dt} \right|_{t=0^+} = 10 \text{ A/s}$, $\left. \frac{di_2}{dt} \right|_{t=0^+} = 5 \text{ A/s}$.

3 Question 3 (Classical Transient Analysis of RC, RL & RLC Circuits)

3.1 2083 Baishakh Exam Paper

Question 3

Derive the expression of current in series RC circuit excited by exponential voltage source by using classical method.

Solution:

1. Circuit Configuration & Differential Equation: Consider a series RC circuit connected to an exponential voltage source $v(t) = V_0 e^{-at}$ for $t \geq 0$. Applying Kirchhoff's Voltage Law (KVL) around the loop:

$$Ri(t) + \frac{1}{C} \int i(t) dt = V_0 e^{-at}$$

Differentiating both sides with respect to t :

$$R \frac{di(t)}{dt} + \frac{1}{C} i(t) = -aV_0 e^{-at}$$

Dividing by R :

$$\frac{di(t)}{dt} + \frac{1}{RC} i(t) = -\frac{aV_0}{R} e^{-at}$$

Let $\alpha = \frac{1}{RC}$ be the inverse time constant parameter. The differential equation becomes:

$$\frac{di(t)}{dt} + \alpha i(t) = -\frac{aV_0}{R} e^{-at}$$

2. Complementary Function ($i_c(t)$): The homogeneous equation is:

$$\frac{di(t)}{dt} + \alpha i(t) = 0$$

The complementary solution is given by:

$$i_c(t) = K e^{-\alpha t} = K e^{-\frac{t}{RC}}$$

3. Particular Integral ($i_p(t)$): Assume a particular solution of the form $i_p(t) = A e^{-at}$. Substitute $i_p(t)$ into the non-homogeneous differential equation:

$$-aA e^{-at} + \alpha A e^{-at} = -\frac{aV_0}{R} e^{-at}$$

$$A(\alpha - a) = -\frac{aV_0}{R} \implies A = \frac{aV_0}{R(a - \alpha)}$$

Thus, the particular integral is:

$$i_p(t) = \frac{aV_0}{R(a - \alpha)} e^{-at}$$

4. Complete Solution & Initial Conditions: The total current is $i(t) = i_c(t) + i_p(t)$:

$$i(t) = Ke^{-\alpha t} + \frac{aV_0}{R(a-\alpha)}e^{-at}$$

Assuming an uncharged capacitor at $t = 0^-$, $v_C(0^+) = 0$. At $t = 0^+$, KVL yields:

$$Ri(0^+) + v_C(0^+) = v(0^+) \implies Ri(0^+) + 0 = V_0 \implies i(0^+) = \frac{V_0}{R}$$

Substituting $t = 0$ into the total current equation:

$$i(0^+) = K + \frac{aV_0}{R(a-\alpha)} = \frac{V_0}{R}$$

$$K = \frac{V_0}{R} - \frac{aV_0}{R(a-\alpha)} = \frac{V_0}{R} \left(1 - \frac{a}{a-\alpha}\right) = \frac{V_0}{R} \left(\frac{-\alpha}{a-\alpha}\right) = \frac{\alpha V_0}{R(\alpha-a)}$$

Substituting K back into the total current equation:

$$i(t) = \frac{\alpha V_0}{R(\alpha-a)}e^{-\alpha t} + \frac{aV_0}{R(a-\alpha)}e^{-at}$$

Since $\alpha = \frac{1}{RC}$:

$$i(t) = \frac{V_0}{R(1-aRC)} \left[e^{-at} - e^{-\frac{t}{RC}} \right] \quad \text{for } a \neq \frac{1}{RC}$$

Final Answer: $i(t) = \frac{V_0}{R(1-aRC)} \left[e^{-at} - e^{-\frac{t}{RC}} \right] \quad \text{for } a \neq \frac{1}{RC}.$

3.2 2082 Bhadra Exam Paper

Question 3

Derive the differential equation of series RLC circuit when excited by step voltage and discuss the different cases of circuit responses.

Solution:

1. Circuit Equation Formulation: Consider a series RLC circuit connected to a step voltage source $V(t) = V_0 u(t)$. Applying KVL to the loop:

$$Ri(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int i(t) dt = V_0$$

Differentiating with respect to t :

$$L \frac{d^2 i(t)}{dt^2} + R \frac{di(t)}{dt} + \frac{1}{C} i(t) = 0$$

Dividing by L :

$$\frac{d^2 i(t)}{dt^2} + \frac{R}{L} \frac{di(t)}{dt} + \frac{1}{LC} i(t) = 0$$

This is the second-order homogeneous linear differential equation for the circuit current.

2. Characteristic Equation and Roots: Assuming a solution $i(t) = K e^{st}$, the characteristic equation is:

$$s^2 + \frac{R}{L} s + \frac{1}{LC} = 0$$

Using standard second-order notation $s^2 + 2\alpha s + \omega_0^2 = 0$:

- **Damping Factor (α):** $\alpha = \frac{R}{2L}$
- **Undamped Resonant Frequency (ω_0):** $\omega_0 = \frac{1}{\sqrt{LC}}$

The roots of the characteristic equation are:

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$$

3. Three Cases of Circuit Responses:

Case 1: Overdamped Response ($\alpha > \omega_0 \implies R > 2\sqrt{\frac{L}{C}}$)

- The roots s_1 and s_2 are real, unequal, and negative.
- **Response Equation:**

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

- **Behavior:** The current slowly rises and decays smoothly to zero without oscillation.

Case 2: Critically Damped Response ($\alpha = \omega_0 \implies R = 2\sqrt{\frac{L}{C}}$)

- The roots are real and equal ($s_1 = s_2 = -\alpha = -\frac{R}{2L}$).

• **Response Equation:**

$$i(t) = (A_1 + A_2 t)e^{-\alpha t}$$

- **Behavior:** The system returns to steady-state in the fastest possible time without oscillating.

Case 3: Underdamped Response ($\alpha < \omega_0 \implies R < 2\sqrt{\frac{L}{C}}$)

- The roots are complex conjugate pairs: $s_{1,2} = -\alpha \pm j\omega_d$, where $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$.

• **Response Equation:**

$$i(t) = e^{-\alpha t} [A_1 \cos(\omega_d t) + A_2 \sin(\omega_d t)]$$

- **Behavior:** The current displays damped sinusoidal oscillations exponentially decaying to zero over time.

Final Answer: Differential Equation: $\frac{d^2 \mathbf{i}}{dt^2} + \frac{R}{L} \frac{d\mathbf{i}}{dt} + \frac{1}{LC} \mathbf{i} = \mathbf{0}$ with Overdamped, Critically Damped, and Underdamped response cases.

3.3 2082 Baishakh Exam Paper

Question 3

A DC source of 100 V is suddenly applied at time $t = 0$ to a series RLC circuit comprising $R = 2 \Omega$, $L = 0.5 \text{ H}$ and $C = 1 \text{ F}$. Obtain the expression for current in the circuit by using classical method.

Solution:

1. Given Parameters:

- $V_0 = 100 \text{ V}$, $R = 2 \Omega$, $L = 0.5 \text{ H}$, $C = 1 \text{ F}$
- Initial conditions: $i(0^+) = 0$, $v_C(0^+) = 0$

2. Differential Equation:

$$L \frac{d^2 i(t)}{dt^2} + R \frac{di(t)}{dt} + \frac{1}{C} i(t) = 0$$

Substituting given values:

$$0.5 \frac{d^2 i}{dt^2} + 2 \frac{di}{dt} + 1i = 0$$

Multiplying by 2:

$$\frac{d^2 i}{dt^2} + 4 \frac{di}{dt} + 2i = 0$$

3. Characteristic Roots:

$$s^2 + 4s + 2 = 0 \implies s = \frac{-4 \pm \sqrt{16 - 8}}{2} = \frac{-4 \pm \sqrt{8}}{2} = -2 \pm \sqrt{2}$$

$$s_1 = -2 + \sqrt{2} \approx -0.586, \quad s_2 = -2 - \sqrt{2} \approx -3.414$$

Since the roots are real and distinct, the circuit response is **overdamped**.

4. General Solution:

$$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} = A_1 e^{(-2+\sqrt{2})t} + A_2 e^{(-2-\sqrt{2})t}$$

5. Applying Initial Conditions:

1. At $t = 0^+$:

$$i(0^+) = A_1 + A_2 = 0 \implies A_2 = -A_1$$

2. Finding $\left. \frac{di}{dt} \right|_{t=0^+}$ from KVL at $t = 0^+$:

$$V_0 = Ri(0^+) + L \left. \frac{di}{dt} \right|_{t=0^+} + v_C(0^+) \implies 100 = 2(0) + 0.5 \left. \frac{di}{dt} \right|_{t=0^+} + 0$$

$$\left. \frac{di}{dt} \right|_{t=0^+} = \frac{100}{0.5} = 200 \text{ A/s}$$

3. Differentiating $i(t)$ at $t = 0^+$:

$$\left. \frac{di}{dt} \right|_{t=0^+} = s_1 A_1 + s_2 A_2 = 200$$

Substitute $A_2 = -A_1$:

$$A_1(s_1 - s_2) = 200 \implies s_1 - s_2 = (-2 + \sqrt{2}) - (-2 - \sqrt{2}) = 2\sqrt{2}$$

$$A_1(2\sqrt{2}) = 200 \implies A_1 = \frac{100}{\sqrt{2}} = 50\sqrt{2} \approx 70.71$$

$$A_2 = -50\sqrt{2} \approx -70.71$$

6. Final Current Expression:

$$i(t) = 50\sqrt{2} \left[e^{(-2+\sqrt{2})t} - e^{(-2-\sqrt{2})t} \right] \text{ A} \quad (t \geq 0)$$

or in decimal form:

$$i(t) = 70.71 \left(e^{-0.586t} - e^{-3.414t} \right) \text{ A}$$

Final Answer: $i(t) = 50\sqrt{2} \left[e^{(-2+\sqrt{2})t} - e^{(-2-\sqrt{2})t} \right] \text{ A} = 70.71 (e^{-0.586t} - e^{-3.414t}) \text{ A}.$
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3.4 2081 Ashwin Exam Paper

Question 3

The circuit shown below is in steady state with switch at position 'a'. The switch is moved to position 'b' at $t = 0$. Find the expression for $V(t)$ and current through 2 H inductor $i_L(t)$ for $t > 0$ using classical method.

Solution:

1. Steady-State Analysis at $t = 0^-$ (Switch at 'a'):

- Switch is at 'a' for $t < 0$. The 1 V DC source is connected to a $1\ \Omega$ resistor and a 1 H inductor.
- In DC steady state, the 1 H inductor acts as a short circuit.
- Initial current through 1 H inductor (i_{L1}):

$$i_{L1}(0^-) = \frac{1\text{ V}}{1\ \Omega} = 1\text{ A}$$

- Initial current through 2 H inductor prior to $t = 0$: $i_{L2}(0^-) = 0\text{ A}$ (disconnected).

By continuity of inductor current:

$$i_{L1}(0^+) = 1\text{ A} \quad (\text{downward through 1 H inductor}), \quad i_{L2}(0^+) = 0\text{ A}$$

2. Circuit Analysis for $t > 0$ (Switch at 'b'): When switched to 'b', the source 1 V and $1\ \Omega$ resistor are isolated. The remaining network consists of:

- Inductor $L_1 = 1\text{ H}$ (with initial current 1 A)
- Resistor $R = 0.5\ \Omega$
- Inductor $L_2 = 2\text{ H}$ (with initial current 0 A)

All three elements are connected in **parallel** across node voltage $V(t)$.

3. Finding Node Voltage $V(t)$: Applying KCL at the top node for $t > 0$:

$$\frac{V(t)}{R} + \frac{1}{L_1} \int_0^t V(\tau) d\tau + i_{L1}(0^+) + \frac{1}{L_2} \int_0^t V(\tau) d\tau + i_{L2}(0^+) = 0$$

Substitute values ($R = 0.5\ \Omega$, $i_{L1}(0^+) = 1\text{ A}$, $i_{L2}(0^+) = 0$):

$$\frac{V(t)}{0.5} + \left(\frac{1}{1} + \frac{1}{2}\right) \int_0^t V(\tau) d\tau + 1 = 0$$

$$2V(t) + \frac{3}{2} \int_0^t V(\tau) d\tau = -1$$

Differentiating both sides with respect to t :

$$2\frac{dV(t)}{dt} + \frac{3}{2}V(t) = 0 \implies \frac{dV(t)}{dt} + \frac{3}{4}V(t) = 0$$

This is a first-order linear differential equation.

$$V(t) = V(0^+)e^{-\frac{3}{4}t}$$

Finding initial voltage $V(0^+)$ at $t = 0^+$:

$$2V(0^+) + 1 = 0 \implies V(0^+) = -0.5 \text{ V}$$

Thus:

$$V(t) = -0.5e^{-0.75t} \text{ V} \quad (t > 0)$$

4. Finding Current through 2 H Inductor $i_{L2}(t)$: Using the $V - I$ relation for $L_2 = 2 \text{ H}$:

$$i_{L2}(t) = i_{L2}(0^+) + \frac{1}{L_2} \int_0^t V(\tau) d\tau$$

Substitute $i_{L2}(0^+) = 0$, $L_2 = 2 \text{ H}$, and $V(\tau) = -0.5e^{-0.75\tau}$:

$$i_{L2}(t) = 0 + \frac{1}{2} \int_0^t (-0.5e^{-0.75\tau}) d\tau = -\frac{0.5}{2} \left[\frac{e^{-0.75\tau}}{-0.75} \right]_0^t = \frac{0.25}{0.75} (e^{-0.75t} - 1)$$

$$i_{L2}(t) = \frac{1}{3} (e^{-0.75t} - 1) \text{ A} = -0.333 (1 - e^{-0.75t}) \text{ A} \quad (t > 0)$$

Final Answer: $V(t) = -0.5e^{-0.75t} \text{ V}$ and $i_{L2}(t) = \frac{1}{3}(e^{-0.75t} - 1) \text{ A} = -0.333(1 - e^{-0.75t}) \text{ A}.$

4 Question 4 (Laplace Transform Applications to Circuit Analysis)

4.1 2083 Baishakh Exam Paper

Question 4

In the given circuit, switch is opened at $t = 0$. Find the current and voltage across inductor using Laplace transform method.

Solution:

1. Initial Conditions ($t = 0^-$): Before $t = 0$, the switch is closed and the circuit is in steady state under DC excitation (4 V):

- Inductor behaves as a **short circuit**.
- Capacitor behaves as an **open circuit**.

Calculating the initial inductor current $i_L(0^-)$:

$$i_L(0^-) = \frac{4 \text{ V}}{1 \Omega} = 4 \text{ A}$$

By continuity of inductor current:

$$i_L(0^+) = i_L(0^-) = 4 \text{ A}$$

2. s -Domain Circuit ($t > 0$): When the switch opens at $t = 0$, the 4 V source and 1 F capacitor are disconnected. The remaining loop consists of the inductor ($L = 0.5 \text{ H}$) and resistor ($R = 1 \Omega$).

In the s -domain:

- Inductor impedance $sL = 0.5s$ in series with initial current voltage source $L \cdot i_L(0^+) = 0.5 \times 4 = 2 \text{ V}$.
- Resistor impedance $R = 1 \Omega$.

Applying KVL in the s -domain loop:

$$(0.5s + 1)I_L(s) = L \cdot i_L(0^+) = 2$$

3. Solving for Inductor Current $i_L(t)$:

$$I_L(s) = \frac{2}{0.5s + 1} = \frac{4}{s + 2}$$

Taking the Inverse Laplace Transform:

$$i_L(t) = \mathcal{L}^{-1} \left\{ \frac{4}{s + 2} \right\} = 4e^{-2t} \text{ A} \quad (t \geq 0)$$

4. Solving for Inductor Voltage $v_L(t)$: Using the inductor relation $v_L(t) = L \frac{di_L(t)}{dt}$:

$$v_L(t) = 0.5 \cdot \frac{d}{dt} (4e^{-2t}) = 0.5 \cdot (-8e^{-2t}) = -4e^{-2t} \text{ V} \quad (t \geq 0)$$

Final Answer: $i_L(t) = 4e^{-2t} \text{ A}$ and $v_L(t) = -4e^{-2t} \text{ V}$.

4.2 2082 Bhadra Exam Paper

Question 4

In the circuit shown switch is changed from position a to position b at $t = 0$. Determine the current $i(t)$ using Laplace transform method.

Solution:

1. Initial Conditions ($t = 0^-$): For $t < 0$, the switch is at position **a** and connected to the 10 V DC source for a long time:

- Inductor acts as a **short circuit**.
- Capacitor acts as an **open circuit**.

Analyzing the circuit at position **a**:

- Initial inductor current: $i_L(0^-) = 0$ A (blocked by capacitor branch).
- Initial capacitor voltage: $v_C(0^-) = 10$ V.

By continuity:

$$i_L(0^+) = 0 \text{ A}, \quad v_C(0^+) = 10 \text{ V}$$

2. s -Domain Circuit ($t > 0$): At $t = 0$, the switch moves to position **b**, placing capacitor ($C = 10 \mu\text{F} = 10^{-5} \text{ F}$) and inductor ($L = 0.1 \text{ H}$) in series loop.

In the s -domain:

- $Z_C(s) = \frac{1}{sC} = \frac{10^5}{s}$
- Initial capacitor voltage source = $\frac{v_C(0^+)}{s} = \frac{10}{s}$
- $Z_L(s) = sL = 0.1s$

Applying KVL in s -domain:

$$\left(0.1s + \frac{10^5}{s}\right) I(s) = \frac{10}{s}$$

3. Solving for $I(s)$ & Inverse Laplace Transform: Multiplying through by s :

$$(0.1s^2 + 10^5)I(s) = 10 \implies I(s) = \frac{10}{0.1s^2 + 10^5} = \frac{100}{s^2 + 10^6}$$

Recognizing standard form $\mathcal{L}^{-1}\left\{\frac{\omega}{s^2 + \omega^2}\right\} = \sin(\omega t)$ where $\omega = \sqrt{10^6} = 1000 \text{ rad/s}$:

$$I(s) = \frac{100}{1000} \cdot \frac{1000}{s^2 + (1000)^2} = 0.1 \cdot \frac{1000}{s^2 + (1000)^2}$$

Taking Inverse Laplace Transform:

$$i(t) = 0.1 \sin(1000t) \text{ A} \quad (t \geq 0)$$

Final Answer: $i(t) = 0.1 \sin(1000t) \text{ A}$.

4.3 2082 Baishakh Exam Paper

Question 4

An exponential voltage $V(t) = 20e^{-4t}$ is suddenly applied at time $t = 0$ to series RLC circuit comprising $R = 2\ \Omega$, $L = 0.5\text{ H}$ and $C = 1\text{ F}$. Obtain the expression for the current $i(t)$ in the circuit using Laplace method.

Solution:

1. Initial Conditions ($t = 0^-$): Assuming zero initial energy stored in reactive components:

$$i(0^-) = 0\text{ A} \implies i(0^+) = 0\text{ A}, \quad v_C(0^-) = 0\text{ V} \implies v_C(0^+) = 0\text{ V}$$

2. s -Domain Formulation:

- Input voltage: $V(t) = 20e^{-4t} \implies V(s) = \frac{20}{s+4}$
- Total impedance $Z(s)$:

$$Z(s) = R + sL + \frac{1}{sC} = 2 + 0.5s + \frac{1}{s} = \frac{0.5s^2 + 2s + 1}{s} = \frac{s^2 + 4s + 2}{2s}$$

The loop current in s -domain:

$$I(s) = \frac{V(s)}{Z(s)} = \frac{\frac{20}{s+4}}{\frac{s^2+4s+2}{2s}} = \frac{40s}{(s+4)(s^2+4s+2)}$$

3. Partial Fraction Expansion: The roots of $s^2 + 4s + 2 = 0$ are $s_{1,2} = -2 \pm \sqrt{2}$. Expanding into partial fractions:

$$\frac{40s}{(s+4)(s^2+4s+2)} = \frac{A}{s+4} + \frac{Bs+C}{s^2+4s+2}$$

1. Finding A :

$$A = \left. \frac{40s}{s^2+4s+2} \right|_{s=-4} = \frac{40(-4)}{(-4)^2+4(-4)+2} = \frac{-160}{16-16+2} = -80$$

2. Finding B and C :

$$40s = A(s^2+4s+2) + (Bs+C)(s+4)$$

$$40s = -80(s^2+4s+2) + Bs^2 + 4Bs + Cs + 4C$$

$$40s = (B-80)s^2 + (4B+C-320)s + (4C-160)$$

Comparing coefficients:

- s^2 coefficient: $B - 80 = 0 \implies \mathbf{B = 80}$
- Constant term: $4C - 160 = 0 \implies \mathbf{C = 40}$

4. Inverse Laplace Transform:

$$I(s) = \frac{-80}{s+4} + \frac{80s+40}{s^2+4s+2}$$

Completing the square in denominator: $s^2 + 4s + 2 = (s+2)^2 - 2$:

$$\frac{80s+40}{(s+2)^2-2} = \frac{80(s+2)-120}{(s+2)^2-2} = 80 \frac{s+2}{(s+2)^2-(\sqrt{2})^2} - \frac{120}{\sqrt{2}} \frac{\sqrt{2}}{(s+2)^2-(\sqrt{2})^2}$$

Taking Inverse Laplace Transform:

$$i(t) = -80e^{-4t} + 80e^{-2t} \cosh(\sqrt{2}t) - 60\sqrt{2}e^{-2t} \sinh(\sqrt{2}t) \text{ A} \quad (t \geq 0)$$

Final Answer: $i(t) = -80e^{-4t} + 80e^{-2t} \cosh(\sqrt{2}t) - 60\sqrt{2}e^{-2t} \sinh(\sqrt{2}t) \text{ A} = -80e^{-4t} + 80e^{-2t} \cosh(\sqrt{2}t) - 84.85e^{-2t} \sinh(\sqrt{2}t) \text{ A}.$

4.4 2081 Ashwin Exam Paper

Question 4

The switch in the figure below has been in position 'a' for a long time. Then, it is moved to 'b' at $t = 0$. Obtain the expression for voltage across capacitor for $t > 0$ using Laplace transform method.

Solution:

1. Initial Conditions ($t = 0^-$): For $t < 0$, the switch is at position **a** connected to a 6 V source for a long time:

- Under DC steady state, the capacitor is an **open circuit** and inductor is a **short circuit**.
- Initial capacitor voltage $v_C(0^-)$: Connected across the 6 V source through the 2Ω resistor. Since no current flows into the open capacitor:

$$v_C(0^-) = 6 \text{ V}$$

- Initial inductor current $i_L(0^-)$: Branch **b** is open for $t < 0$, so no current flows:

$$i_L(0^-) = 0 \text{ A}$$

By continuity:

$$v_C(0^+) = 6 \text{ V}, \quad i_L(0^+) = 0 \text{ A}$$

2. s-Domain Circuit ($t > 0$): At $t = 0$, the switch moves to position **b**. The 6 V supply and 2Ω resistor are isolated. The remaining loop consists of 1Ω resistor, 0.5 H inductor, and 1 F capacitor in series.

In the s -domain:

- Impedances: $R = 1\Omega$, $sL = 0.5s$, $\frac{1}{sC} = \frac{1}{s}$
- Initial capacitor voltage source: $\frac{v_C(0^+)}{s} = \frac{6}{s}$

Applying KVL in s -domain:

$$\left(1 + 0.5s + \frac{1}{s}\right) I(s) = \frac{6}{s}$$

3. Solving for Loop Current $I(s)$ & Capacitor Voltage $V_C(s)$: Multiplying through by s :

$$(0.5s^2 + s + 1)I(s) = 6 \implies I(s) = \frac{6}{0.5s^2 + s + 1} = \frac{12}{s^2 + 2s + 2}$$

The voltage across the capacitor in the s -domain:

$$V_C(s) = \frac{v_C(0^+)}{s} - I(s) \cdot \frac{1}{sC} = \frac{6}{s} - \frac{12}{s(s^2 + 2s + 2)}$$

Decomposing $\frac{12}{s(s^2 + 2s + 2)}$ into partial fractions:

$$\frac{12}{s(s^2 + 2s + 2)} = \frac{K_1}{s} + \frac{K_2s + K_3}{s^2 + 2s + 2}$$

$$1. K_1 = \frac{12}{s^2+2s+2} \Big|_{s=0} = \frac{12}{2} = 6$$

$$2. 12 = 6(s^2 + 2s + 2) + (K_2s + K_3)s = (6 + K_2)s^2 + (12 + K_3)s + 12$$

$$6 + K_2 = 0 \implies K_2 = -6, \quad 12 + K_3 = 0 \implies K_3 = -12$$

Substituting back into $V_C(s)$:

$$V_C(s) = \frac{6}{s} - \left(\frac{6}{s} - \frac{6s + 12}{s^2 + 2s + 2} \right) = \frac{6s + 12}{s^2 + 2s + 2}$$

4. Inverse Laplace Transform: Completing the square in the denominator: $s^2 + 2s + 2 = (s + 1)^2 + 1^2$:

$$V_C(s) = \frac{6(s + 1) + 6}{(s + 1)^2 + 1} = 6 \frac{s + 1}{(s + 1)^2 + 1^2} + 6 \frac{1}{(s + 1)^2 + 1^2}$$

Taking Inverse Laplace Transform:

$$v_C(t) = 6e^{-t} \cos(t) + 6e^{-t} \sin(t) \text{ V } (t > 0)$$

Final Answer: $v_C(t) = 6e^{-t} (\cos(t) + \sin(t)) \text{ V}.$

5 Question 5 (Two-Port Networks & Bode Plots)

5.1 2083 Baishakh Exam Paper

Question 5

What is network function? Plot Asymptotic Bode graph of given transfer function:

$$H(s) = \frac{10(s^2 + 8s + 15)}{s^2(s + 3)(0.05s + 1)}$$

Solution:

1. Definition of Network Function: A network function $H(s)$ is defined as the ratio of the Laplace transform of the output signal $Y(s)$ (response) to the Laplace transform of the input signal $X(s)$ (excitation) under initial conditions set to zero:

$$H(s) = \frac{Y(s)}{X(s)}$$

It characterizes the LTI system's dynamic behavior, frequency response, stability, and transfer characteristics.

2. Simplifying $H(s)$ to Standard Time-Constant (Bode) Form: Factoring the numerator quadratic:

$$s^2 + 8s + 15 = (s + 3)(s + 5)$$

Substitute into $H(s)$:

$$H(s) = \frac{10(s + 3)(s + 5)}{s^2(s + 3)(0.05s + 1)}$$

Canceling the common pole-zero term $(s + 3)$:

$$H(s) = \frac{10(s + 5)}{s^2(0.05s + 1)}$$

Converting numerator to standard form $(1 + s/\omega_z)$:

$$10(s + 5) = 10 \cdot 5 \left(1 + \frac{s}{5}\right) = 50 \left(1 + \frac{s}{5}\right)$$

Thus:

$$H(s) = \frac{50 \left(1 + \frac{s}{5}\right)}{s^2(1 + 0.05s)} = \frac{50 \left(1 + \frac{s}{5}\right)}{s^2 \left(1 + \frac{s}{20}\right)}$$

3. Bode Magnitude Parameters:

- **System Gain $K = 50$:** $20 \log_{10}(50) \approx 34$ dB.
- **Pole at Origin (s^2):** Double pole at origin gives an initial low-frequency slope of -40 dB/dec.
- **Corner Frequencies:**
 - $\omega_{z1} = 5$ rad/s (Simple Zero at $s = -5$)
 - $\omega_{p1} = 20$ rad/s (Simple Pole at $s = -20$)

4. Bode Magnitude Slope Table:

Frequency Range (rad/s)	Factor	Change (dB/dec)	Total Slope (dB/dec)
$\omega < 5$	$\frac{50}{s^2}$	—	—40
$5 \leq \omega < 20$	Zero at $\omega = 5$	+20	$-40 + 20 = -20$
$\omega \geq 20$	Pole at $\omega = 20$	-20	$-20 - 20 = -40$

5. Magnitude Reference Points:

- At $\omega = 1$ rad/s: $|H(j1)|_{\text{dB}} = 20 \log_{10}(50) - 20 \log_{10}(1^2) = 34$ dB.
- At $\omega = 5$ rad/s: $|H(j5)|_{\text{dB}} = 34 - 40 \log_{10}(5) = 34 - 27.96 = \mathbf{6.04}$ dB.
- At $\omega = 20$ rad/s: $|H(j20)|_{\text{dB}} = 6.04 - 20 \log_{10}\left(\frac{20}{5}\right) = 6.04 - 12.04 = \mathbf{-6.00}$ dB.

Final Answer: Standard Form: $H(s) = \frac{50(1+s/5)}{s^2(1+s/20)}$. Slopes: -40 dB/dec ($\omega < 5$), -20 dB/dec ($5 \leq \omega < 20$), -40 dB/dec ($\omega \geq 20$).

5.2 2082 Bhadra Exam Paper

Question 5

Define two port network. For the given network containing a series branch 3Ω , dependent voltage source $2I_a$, $-j2\Omega$ capacitor branch, 1Ω resistor carrying current I_a , and $j5\Omega$ inductor, find the Z parameters.

Solution:

1. Definition of Two-Port Network: A **two-port network** is an electrical network with two separate pairs of terminals (ports): Port 1 (input) and Port 2 (output). The current entering one terminal of a port is equal to the current leaving the other terminal of the same port (I_1 at Port 1 and I_2 at Port 2).

2. Z-Parameter Governing Equations:

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

From the network configuration:

- Port 1 current I_1 enters from the left.
- Port 2 current I_2 enters from the right.
- Current flowing right-to-left through the $1\Omega + j5\Omega$ branch is I_2 , so $I_a = -I_2$.
- Total current through the middle shunt capacitor branch ($-j2\Omega$) is $(I_1 + I_2)$.

3. Deriving Z-Parameters via KVL:

A. KVL at Port 1 Loop:

$$V_1 = 3I_1 + 2I_a - j2(I_1 + I_2)$$

Substitute $I_a = -I_2$:

$$V_1 = 3I_1 + 2(-I_2) - j2I_1 - j2I_2 = (3 - j2)I_1 + (-2 - j2)I_2$$

Comparing coefficients with $V_1 = Z_{11}I_1 + Z_{12}I_2$:

$$Z_{11} = (3 - j2)\Omega, \quad Z_{12} = (-2 - j2)\Omega$$

B. KVL at Port 2 Loop: Starting from Port 2:

$$V_2 = (1 + j5)I_2 - 2I_a - j2(I_1 + I_2)$$

Substitute $I_a = -I_2$:

$$V_2 = (1 + j5)I_2 - 2(-I_2) - j2I_1 - j2I_2 = -j2I_1 + (1 + j5 + 2 - j2)I_2$$

$$V_2 = -j2I_1 + (3 + j3)I_2$$

Comparing coefficients with $V_2 = Z_{21}I_1 + Z_{22}I_2$:

$$Z_{21} = -j2\Omega, \quad Z_{22} = (3 + j3)\Omega$$

4. Final Z-Parameter Matrix:

$$[Z] = \begin{bmatrix} 3 - j2 & -2 - j2 \\ -j2 & 3 + j3 \end{bmatrix} \Omega$$

Final Answer: $[Z] = \begin{bmatrix} 3 - j2 & -2 - j2 \\ -j2 & 3 + j3 \end{bmatrix} \Omega.$

5.3 2082 Baishakh Exam Paper

Question 5

Sketch the asymptotic Bode plot for the transfer function given by:

$$T(s) = \frac{100s(s+5)}{(s+1)(s^2+10s+100)}$$

Solution:

1. Convert $T(s)$ into Standard Bode Form: Factoring out constants from numerator and denominator:

- Numerator zero at $s = -5$: $(s+5) = 5\left(1 + \frac{s}{5}\right)$
- Denominator quadratic term $s^2 + 10s + 100$: $100\left(1 + \frac{s}{10} + \frac{s^2}{100}\right)$

Substitute into $T(s)$:

$$T(s) = \frac{100 \cdot s \cdot 5 \left(1 + \frac{s}{5}\right)}{1 \cdot (1+s) \cdot 100 \left(1 + \frac{s}{10} + \frac{s^2}{100}\right)} = \frac{5s \left(1 + \frac{s}{5}\right)}{(1+s) \left(1 + \frac{s}{10} + \frac{s^2}{100}\right)}$$

2. System Parameters:

- **System Gain $K = 5$:** $20 \log_{10}(5) \approx 14$ dB.
- **Single Zero at Origin (s):** Initial low-frequency slope of $+20$ dB/dec.
- **Pole at $\omega_{p1} = 1$ rad/s:** Slope change -20 dB/dec.
- **Zero at $\omega_{z1} = 5$ rad/s:** Slope change $+20$ dB/dec.
- **Complex Conjugate Poles at $\omega_n = \sqrt{100} = 10$ rad/s:** Slope change -40 dB/dec.

3. Bode Magnitude Slope Table:

Frequency Range (rad/s)	Factor	Change (dB/dec)	Total Slope (dB/dec)
$\omega < 1$	$5s$	—	$+20$
$1 \leq \omega < 5$	Pole at $\omega = 1$	-20	$+20 - 20 = \mathbf{0}$
$5 \leq \omega < 10$	Zero at $\omega = 5$	$+20$	$0 + 20 = \mathbf{+20}$
$\omega \geq 10$	Complex poles at $\omega = 10$	-40	$+20 - 40 = \mathbf{-20}$

4. Magnitude Calculations at Corner Frequencies:

- At $\omega = 1$ rad/s: $|T(j1)|_{\text{dB}} = 20 \log_{10}(5 \cdot 1) = \mathbf{14}$ dB.
- At $\omega = 5$ rad/s: Slope is 0 dB/dec from $\omega = 1$ to 5 , so $|T(j5)|_{\text{dB}} = \mathbf{14}$ dB.

- At $\omega = 10$ rad/s: Slope is +20 dB/dec from $\omega = 5$ to 10:

$$|T(j10)|_{\text{dB}} = 14 + 20 \log_{10} \left(\frac{10}{5} \right) = 14 + 6.02 = \mathbf{20.02 \text{ dB}}$$

Final Answer: Standard Form: $T(s) = \frac{5s(1+s/5)}{(1+s)(1+0.1s+0.01s^2)}$. Corner frequencies at 1, 5, 10 rad/s with asymptotic magnitudes 14 dB, 14 dB, 20.02 dB.

5.4 2081 Ashwin Exam Paper

Question 5

Plot asymptotic Bode graph of given transfer function:

$$G(s) = \frac{500(s+2)}{s(s^2+15s+121)}$$

Solution:

1. Convert $G(s)$ to Standard Time-Constant Form: Factoring constants:

- Numerator: $500(s+2) = 500 \cdot 2 \left(1 + \frac{s}{2}\right) = 1000 \left(1 + \frac{s}{2}\right)$
- Denominator quadratic: $s^2 + 15s + 121 = 121 \left(1 + \frac{15}{121}s + \frac{s^2}{121}\right)$

Substitute into $G(s)$:

$$G(s) = \frac{1000 \left(1 + \frac{s}{2}\right)}{121 \cdot s \left(1 + \frac{15}{121}s + \frac{s^2}{121}\right)} = \frac{8.264 \left(1 + \frac{s}{2}\right)}{s \left(1 + \frac{15}{121}s + \frac{s^2}{121}\right)}$$

2. Critical Parameters:

- **System Gain $K = 8.264$:** $20 \log_{10}(8.264) \approx 18.34$ dB.
- **Pole at Origin (s):** Initial low-frequency slope of -20 dB/dec.
- **Zero at $\omega_{z1} = 2$ rad/s:** Slope change $+20$ dB/dec.
- **Complex Poles at $\omega_n = \sqrt{121} = 11$ rad/s:** Slope change -40 dB/dec.

3. Bode Magnitude Slope Table:

Frequency Range (rad/s)	Factor	Change (dB/dec)	Total Slope (dB/dec)
$\omega < 2$	$\frac{8.264}{s}$	—	-20
$2 \leq \omega < 11$	Zero at $\omega = 2$	$+20$	$-20 + 20 = 0$
$\omega \geq 11$	Complex poles at $\omega = 11$	-40	$0 - 40 = -40$

4. Magnitude Calculations at Key Frequencies:

- At $\omega = 1$ rad/s: $|G(j1)|_{\text{dB}} = 20 \log_{10}(8.264) = \mathbf{18.34}$ dB.
- At $\omega = 2$ rad/s: $|G(j2)|_{\text{dB}} = 18.34 - 20 \log_{10}(2) = 18.34 - 6.02 = \mathbf{12.32}$ dB.
- At $\omega = 11$ rad/s: Slope is 0 dB/dec from $\omega = 2$ to 11 , so $|G(j11)|_{\text{dB}} = \mathbf{12.32}$ dB.

Final Answer: Standard Form: $G(s) = \frac{8.264(1+s/2)}{s(1+\frac{15}{121}s+\frac{s^2}{121})}$. Slopes: -20 dB/dec ($\omega < 2$), 0 dB/dec ($2 \leq \omega < 11$), -40 dB/dec ($\omega \geq 11$).

6 Question 6 (Two-Port Parameters & Frequency Response / Bode Plots)

6.1 2083 Baishakh Exam Paper

Question 6

Find Z parameter for given two port network and state whether network is symmetrical or reciprocal.

Solution:

1. Network Configuration:

- Series resistor at Port 1: 8Ω
- Shunt resistor across middle branch: 20Ω
- Series resistor between middle and Port 2: 10Ω
- Shunt resistor across Port 2: 20Ω

2. Finding Z_{11} and Z_{21} (Port 2 Open: $I_2 = 0$): When $I_2 = 0$, the 20Ω resistor at Port 2 is in series with the 10Ω resistor.

1. **Input Impedance** $Z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0}$: Equivalent resistance seen from Port 1:

$$R_{eq1} = 8 + [20 \parallel (10 + 20)] = 8 + [20 \parallel 30]$$

$$20 \parallel 30 = \frac{20 \times 30}{20 + 30} = 12\Omega \implies Z_{11} = 8 + 12 = \mathbf{20\Omega}$$

2. **Transfer Impedance** $Z_{21} = \left. \frac{V_2}{I_1} \right|_{I_2=0}$: Voltage across middle 20Ω resistor is $V_m = I_1 \times (20 \parallel 30) = 12I_1$. By voltage division across $(10 + 20)\Omega$ branch:

$$V_2 = V_m \times \frac{20}{10 + 20} = (12I_1) \times \frac{20}{30} = 8I_1 \implies Z_{21} = \frac{V_2}{I_1} = \mathbf{8\Omega}$$

3. Finding Z_{22} and Z_{12} (Port 1 Open: $I_1 = 0$): When $I_1 = 0$, the 8Ω resistor is open and carries no current.

1. **Output Impedance** $Z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0}$: Equivalent resistance seen from Port 2:

$$R_{eq2} = 20 \parallel (10 + 20) = 20 \parallel 30 = \mathbf{12\Omega} \implies Z_{22} = \mathbf{12\Omega}$$

2. **Transfer Impedance** $Z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1=0}$: By current division, current entering $(10 + 20)\Omega$ branch from I_2 :

$$I_{\text{branch}} = I_2 \times \frac{20}{20 + (10 + 20)} = 0.4I_2$$

Voltage V_1 equals voltage across middle 20Ω resistor:

$$V_1 = I_{\text{branch}} \times 20 = (0.4I_2) \times 20 = 8I_2 \implies Z_{12} = \frac{V_1}{I_2} = \mathbf{8\Omega}$$

4. Impedance Matrix & Network Properties:

$$[Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} 20 & 8 \\ 8 & 12 \end{bmatrix} \Omega$$

- **Reciprocity Condition:** $Z_{12} = Z_{21} = 8 \Omega \implies$ Network is **Reciprocal**.
- **Symmetry Condition:** $Z_{11} = 20 \Omega \neq 12 \Omega = Z_{22} \implies$ Network is **Not Symmetrical**.

Final Answer: $[Z] = \begin{bmatrix} 20 & 8 \\ 8 & 12 \end{bmatrix} \Omega$. Network is **Reciprocal** ($Z_{12} = Z_{21}$) but **Not Symmetrical** ($Z_{11} \neq Z_{22}$).

6.2 2082 Bhadra Exam Paper

Question 6

Define poles and zeros of network function. Construct the asymptotic magnitude bode plot and phase plot for given transfer function:

$$T(s) = \frac{100(s+5)}{s^2(s^2+21s+20)}$$

Solution:

Part A: Definition of Poles and Zeros

- **Poles:** Values of s for which the network function $T(s) \rightarrow \infty$. They are the roots of the denominator polynomial.
- **Zeros:** Values of s for which the network function $T(s) \rightarrow 0$. They are the roots of the numerator polynomial.

Part B: Bode Plot Construction

1. Factorization into Standard Bode Form: Factoring denominator $s^2 + 21s + 20 = (s+1)(s+20)$:

$$T(s) = \frac{100(s+5)}{s^2(s+1)(s+20)} = \frac{100 \times 5 \left(1 + \frac{s}{5}\right)}{s^2 \times 1(1+s) \times 20 \left(1 + \frac{s}{20}\right)} = \frac{25 \left(1 + \frac{s}{5}\right)}{s^2(1+s) \left(1 + \frac{s}{20}\right)}$$

System Gain $K = 25 \Rightarrow K_{dB} = 20 \log_{10}(25) \approx 27.96 \text{ dB}$.

2. Identification of Critical Frequencies:

- Double pole at origin (s^2): Initial low-frequency slope -40 dB/dec .
- Corner Frequencies in ascending order: $\omega_{p1} = 1 \text{ rad/s} < \omega_{z1} = 5 \text{ rad/s} < \omega_{p2} = 20 \text{ rad/s}$.

3. Bode Magnitude Slope Table:

Frequency Range (rad/s)	Factor	Change (dB/dec)	Total Slope (dB/dec)
$\omega < 1$	$K = 25, \frac{1}{s^2}$	—	-40
$1 \leq \omega < 5$	Pole at $\omega = 1$	-20	$-40 - 20 = -60$
$5 \leq \omega < 20$	Zero at $\omega = 5$	$+20$	$-60 + 20 = -40$
$\omega \geq 20$	Pole at $\omega = 20$	-20	$-40 - 20 = -60$

4. Key Magnitude Calculations:

- At $\omega = 1 \text{ rad/s}$: $|T(j1)|_{dB} = 20 \log_{10}(25) - 40 \log_{10}(1) = \mathbf{27.96 \text{ dB}}$.
- At $\omega = 5 \text{ rad/s}$: $|T(j5)|_{dB} = 27.96 - 60 \log_{10}(5) = 27.96 - 41.93 = \mathbf{-13.97 \text{ dB}}$.
- At $\omega = 20 \text{ rad/s}$: $|T(j20)|_{dB} = -13.97 - 40 \log_{10}(4) = -13.97 - 24.08 = \mathbf{-38.05 \text{ dB}}$.

5. Phase Angle Function $\phi(\omega)$:

$$\phi(\omega) = -180^\circ + \tan^{-1}\left(\frac{\omega}{5}\right) - \tan^{-1}\left(\frac{\omega}{1}\right) - \tan^{-1}\left(\frac{\omega}{20}\right)$$

- $\omega \rightarrow 0$: $\phi = -180^\circ$
- $\omega = 1$: $\phi = -180^\circ + 11.31^\circ - 45^\circ - 2.86^\circ = -\mathbf{216.55^\circ}$
- $\omega = 5$: $\phi = -180^\circ + 45^\circ - 78.69^\circ - 14.04^\circ = -\mathbf{227.73^\circ}$
- $\omega = 20$: $\phi = -180^\circ + 75.96^\circ - 87.14^\circ - 45^\circ = -\mathbf{236.18^\circ}$
- $\omega \rightarrow \infty$: $\phi = -180^\circ + 90^\circ - 90^\circ - 90^\circ = -\mathbf{270^\circ}$

Final Answer: Bode Form: $T(s) = \frac{25(1+s/5)}{s^2(1+s)(1+s/20)}$. Slopes: $-40, -60, -40, -60$ dB/dec. Phase ranges from -180° to -270° .
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6.3 2082 Baishakh Exam Paper

Question 6

Find the expression of hybrid parameters in terms of Z parameters. Derive the equivalent T parameters of two cascade connected two port networks.

Solution:

Part A: Expressing H-parameters in terms of Z-parameters

1. Governing Equations:

- Z-parameter equations:

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \quad \text{--- (1)}, \quad V_2 = Z_{21}I_1 + Z_{22}I_2 \quad \text{--- (2)}$$

- H-parameter equations:

$$V_1 = h_{11}I_1 + h_{12}V_2 \quad \text{--- (3)}, \quad I_2 = h_{21}I_1 + h_{22}V_2 \quad \text{--- (4)}$$

2. Derivation: From Z-equation (2), express I_2 in terms of I_1 and V_2 :

$$V_2 = Z_{21}I_1 + Z_{22}I_2 \implies Z_{22}I_2 = -Z_{21}I_1 + V_2 \implies I_2 = \left(-\frac{Z_{21}}{Z_{22}}\right)I_1 + \left(\frac{1}{Z_{22}}\right)V_2 \quad \text{--- (5)}$$

Comparing equation (5) with H-equation (4):

$$h_{21} = -\frac{Z_{21}}{Z_{22}}, \quad h_{22} = \frac{1}{Z_{22}}$$

Substitute I_2 from (5) into Z-equation (1):

$$\begin{aligned} V_1 &= Z_{11}I_1 + Z_{12}\left(-\frac{Z_{21}}{Z_{22}}I_1 + \frac{1}{Z_{22}}V_2\right) = \left(Z_{11} - \frac{Z_{12}Z_{21}}{Z_{22}}\right)I_1 + \left(\frac{Z_{12}}{Z_{22}}\right)V_2 \\ V_1 &= \left(\frac{\Delta_Z}{Z_{22}}\right)I_1 + \left(\frac{Z_{12}}{Z_{22}}\right)V_2 \end{aligned}$$

Comparing with H-equation (3):

$$h_{11} = \frac{\Delta_Z}{Z_{22}}, \quad h_{12} = \frac{Z_{12}}{Z_{22}} \quad (\text{where } \Delta_Z = Z_{11}Z_{22} - Z_{12}Z_{21})$$

Part B: Equivalent Transmission (ABCD) Parameters for Cascaded Networks

1. Network Equations: For two networks N_A and N_B connected in cascade:

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A_A & B_A \\ C_A & D_A \end{bmatrix} \begin{bmatrix} V_2' \\ -I_2' \end{bmatrix}, \quad \begin{bmatrix} V_1'' \\ I_1'' \end{bmatrix} = \begin{bmatrix} A_B & B_B \\ C_B & D_B \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

2. Cascade Interconnection Conditions: At the junction between N_A and N_B :

$$V_2' = V_1'' \quad \text{and} \quad -I_2' = I_1''$$

3. Matrix Multiplication:

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A_A & B_A \\ C_A & D_A \end{bmatrix} \begin{bmatrix} A_B & B_B \\ C_B & D_B \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

Thus, the equivalent Transmission matrix $\mathbf{T}_{\text{eq}}$ is:

$$\mathbf{T}_{\text{eq}} = \mathbf{T}_A \times \mathbf{T}_B = \begin{bmatrix} A_A A_B + B_A C_B & A_A B_B + B_A D_B \\ C_A A_B + D_A C_B & C_A B_B + D_A D_B \end{bmatrix}$$

Final Answer: $[H] = \begin{bmatrix} \frac{\Delta_Z}{Z_{22}} & \frac{Z_{12}}{Z_{22}} \\ -\frac{Z_{21}}{Z_{22}} & \frac{1}{Z_{22}} \end{bmatrix}$ and $\mathbf{T}_{\text{eq}} = \mathbf{T}_A \times \mathbf{T}_B$.

6.4 2081 Ashwin Exam Paper

Question 6

Find the expression of Z parameters in terms of $ABCD$ parameters. For given two port networks containing series 8Ω resistor, dependent current source $0.5I_x$, and shunt 5Ω resistor carrying current I_x , find $ABCD$ parameters.

Solution:

Part A: Expressing Z-parameters in terms of ABCD parameters

1. Governing Equations:

- ABCD parameter equations:

$$V_1 = AV_2 - BI_2 \quad \text{--- (1)}, \quad I_1 = CV_2 - DI_2 \quad \text{--- (2)}$$

- Z parameter equations:

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \quad \text{--- (3)}, \quad V_2 = Z_{21}I_1 + Z_{22}I_2 \quad \text{--- (4)}$$

- ##### 2. Derivation:
- From equation (2), express V_2 in terms of I_1 and I_2 :

$$CV_2 = I_1 + DI_2 \implies V_2 = \left(\frac{1}{C}\right)I_1 + \left(\frac{D}{C}\right)I_2 \quad \text{--- (5)}$$

Comparing equation (5) with Z-equation (4):

$$Z_{21} = \frac{1}{C}, \quad Z_{22} = \frac{D}{C}$$

Substitute V_2 from (5) into equation (1):

$$V_1 = A\left(\frac{1}{C}I_1 + \frac{D}{C}I_2\right) - BI_2 = \left(\frac{A}{C}\right)I_1 + \left(\frac{AD - BC}{C}\right)I_2$$

Comparing with Z-equation (3):

$$Z_{11} = \frac{A}{C}, \quad Z_{12} = \frac{\Delta_T}{C} = \frac{AD - BC}{C}$$

Result matrix:

$$[Z] = \begin{bmatrix} \frac{A}{C} & \frac{AD-BC}{C} \\ \frac{1}{C} & \frac{D}{C} \end{bmatrix}$$

Part B: Finding ABCD Parameters for the Given Circuit

1. Circuit Relations:

- Controlling current through 5Ω resistor: $I_x = \frac{V_2}{5}$.
- Dependent current source at Port 1: $0.5I_x = 0.5\left(\frac{V_2}{5}\right) = 0.1V_2$.

2. Finding $A = \left. \frac{V_1}{V_2} \right|_{I_2=0}$ **and** $C = \left. \frac{I_1}{V_2} \right|_{I_2=0}$: When $I_2 = 0$, open-circuit at Port 2 gives $I_x = 0 \implies 0.5I_x = 0$.

- No current flows through 8Ω series resistor, so $V_1 = V_2 \implies \mathbf{A} = \mathbf{1}$.
- KCL at Port 1 node: $I_1 = 0.5I_x + I_{\text{series}} = 0 + 0 = 0 \implies \mathbf{C} = \mathbf{0\,S}$.

3. Finding $B = \left. \frac{V_1}{-I_2} \right|_{V_2=0}$ **and** $D = \left. \frac{I_1}{-I_2} \right|_{V_2=0}$: When Port 2 is short-circuited ($V_2 = 0$):

- $I_x = \frac{V_2}{5} = 0 \implies$ Dependent source $0.5I_x = 0$.
- Current $-I_2$ flows through 8Ω series resistor: $V_1 = 8 \cdot (-I_2) \implies \mathbf{B} = \mathbf{8\,\Omega}$.
- KCL at node 1 gives $I_1 = -I_2 \implies \mathbf{D} = \mathbf{1}$.

Final Answer: $[Z] = \begin{bmatrix} \frac{A}{C} & \frac{AD-BC}{C} \\ \frac{1}{C} & \frac{D}{C} \end{bmatrix}$ and $\mathbf{T} = \begin{bmatrix} 1 & 8\,\Omega \\ 0\,\text{S} & 1 \end{bmatrix}$.

7 Question 7 (Transformers & DC Machines)

7.1 2083 Baishakh Exam Paper

Question 7

List out features of ideal transformer. Show that main magnetic flux in the transformer core is independent of load current.

Solution:

1. Features of an Ideal Transformer:

- **Zero Winding Resistance:** Primary and secondary winding resistances are zero ($R_1 = 0, R_2 = 0$), resulting in zero copper loss ($I^2R = 0$).
- **Infinite Core Permeability ($\mu \rightarrow \infty$):** Negligible magnetizing current is required to establish the required core flux.
- **Zero Leakage Flux:** All magnetic flux links both primary and secondary windings (100% magnetic coupling).
- **Zero Core Loss:** No hysteresis or eddy current losses in the magnetic core.
- **100% Efficiency:** Input power equals output power ($V_1 I_1 = V_2 I_2$).

2. Proof: Main Magnetic Flux in Core is Independent of Load Current:

1. **At No-Load Condition:** When an ideal transformer is excited by a sinusoidal supply voltage V_1 , the primary back EMF E_1 balances V_1 :

$$V_1 = E_1 = 4.44fN_1\phi_m \implies \phi_m = \frac{V_1}{4.44fN_1}$$

Since supply voltage V_1 , supply frequency f , and turns N_1 are constant, the main core flux ϕ_m remains constant.

2. At Loaded Condition:

- Connecting a load to secondary causes secondary current I_2 to flow.
- This produces a demagnetizing MMF ($N_2 I_2$), creating secondary flux ϕ_2 opposing ϕ_m (Lenz's Law).
- The net core flux temporarily decreases, which reduces primary induced EMF E_1 .
- The potential difference ($V_1 - E_1$) causes primary winding to draw an additional current I'_2 (load component of primary current) from the source such that:

$$N_1 I'_2 = N_2 I_2$$

- I'_2 produces a flux ϕ'_2 equal and opposite to ϕ_2 ($\phi'_2 = \phi_2$), completely neutralizing the demagnetizing effect.

3. **Conclusion:** The demagnetizing action of load current I_2 is immediately countered by the additional primary current I'_2 . Hence, the net core flux remains constant at ϕ_m regardless of load current.

Final Answer: Ideal Transformer: Zero losses, infinite permeability, 100% efficiency. Core flux $\phi_m = \frac{V_1}{4.44fN_1}$ is constant because secondary demagnetizing MMF (N_2I_2) is neutralized by primary load MMF ($N_1I'_2$).

7.2 2082 Bhadra Exam Paper

Question 7

State the role of commutator and carbon brushes in DC machine. Explain the characteristics of DC generators.

Solution:

1. Role of Commutator and Carbon Brushes:

• Commutator:

- **In DC Generator:** Functions as a mechanical rectifier converting alternating voltage (AC) induced in rotating armature conductors into direct voltage (DC) at external terminals.
- **In DC Motor:** Reverses current direction in armature conductors every half-cycle to produce continuous unidirectional electromagnetic torque.

• Carbon Brushes:

- Act as sliding electrical contacts to collect current from (or feed current into) rotating commutator segments.
- Made of carbon/graphite due to self-lubricating properties, negative temperature coefficient of resistance (reduces sparking), and lower hardness (prevents wear on copper commutator).

2. Characteristics of DC Generators:

1. Open Circuit Characteristic (OCC) / Magnetization Curve [E_0 vs. I_f]:

- Relation between generated EMF at no-load (E_0) and field current (I_f) at constant speed.
- Starts from residual voltage at $I_f = 0$ due to residual magnetism, increases linearly, and bends due to magnetic core saturation.

2. Internal Characteristic [E vs. I_a]:

- Relation between actual generated EMF under load (E) and armature current (I_a).
- E is lower than E_0 due to demagnetizing armature reaction.

3. External Characteristic [V vs. I_L]:

- Relation between terminal voltage (V) and load current (I_L).
- V is lower than E due to internal armature resistance drop ($I_a R_a$).
- For Shunt Generator: Voltage drops slightly with load.
- For Series Generator: Voltage rises initially with load and falls under heavy load due to saturation and armature reaction.

Final Answer: Commutator: Mechanical rectifier ($AC \leftrightarrow DC$). Brushes: Sliding contacts. Generator curves: OCC (E_0 vs I_f), Internal (E vs I_a), External (V vs I_L).

7.3 2082 Baishakh Exam Paper

Question 7

The following test results were obtained on 20 kVA, 2200/220 V, 50 Hz, single phase transformer:

- **O.C. test:** 220 V, 1.1 A, 125 W (On LV side)
- **S.C. test:** 52.7 V, 8.4 A, 287 W (On HV side)

Calculate the equivalent circuit parameters referred to secondary side.

Solution:

1. Given Data & Transformation Ratio:

- Rating: 20 kVA, 2200/220 V, 50 Hz
- Primary (HV) side = 2200 V, Secondary (LV) side = 220 V
- Transformation ratio $k = \frac{V_2}{V_1} = \frac{220}{2200} = 0.1$

2. Open Circuit (O.C.) Test Calculations (LV / Secondary Side): Measured on LV side: $V_{02} = 220$ V, $I_{02} = 1.1$ A, $W_0 = 125$ W.

1. No-Load Power Factor ($\cos \phi_0$):

$$W_0 = V_{02} I_{02} \cos \phi_0 \implies \cos \phi_0 = \frac{125}{220 \times 1.1} = \frac{125}{242} \approx 0.5165$$

$$\sin \phi_0 = \sqrt{1 - 0.5165^2} \approx 0.8563$$

2. Core Loss Current (I_w) and Magnetizing Current (I_m):

$$I_w = I_{02} \cos \phi_0 = 1.1 \times 0.5165 = 0.568 \text{ A}$$

$$I_m = I_{02} \sin \phi_0 = 1.1 \times 0.8563 = 0.942 \text{ A}$$

3. Shunt Branch Parameters referred to Secondary (LV) Side:

$$R_{c2} = \frac{V_{02}}{I_w} = \frac{220}{0.568} = \mathbf{387.32 \Omega}$$

$$X_{m2} = \frac{V_{02}}{I_m} = \frac{220}{0.942} = \mathbf{233.55 \Omega}$$

3. Short Circuit (S.C.) Test Calculations (HV / Primary Side): Measured on HV side: $V_{sc1} = 52.7 \text{ V}$, $I_{sc1} = 8.4 \text{ A}$, $W_{sc} = 287 \text{ W}$.

1. Equivalent Parameters referred to Primary (HV) Side:

$$R_{e1} = \frac{W_{sc}}{I_{sc1}^2} = \frac{287}{(8.4)^2} = \frac{287}{70.56} = 4.067 \Omega$$

$$Z_{e1} = \frac{V_{sc1}}{I_{sc1}} = \frac{52.7}{8.4} = 6.274 \Omega$$

$$X_{e1} = \sqrt{Z_{e1}^2 - R_{e1}^2} = \sqrt{(6.274)^2 - (4.067)^2} = \sqrt{39.36 - 16.54} = 4.777 \Omega$$

2. Referring Equivalent Parameters to Secondary (LV) Side:

$$R_{e2} = k^2 R_{e1} = (0.1)^2 \times 4.067 = 0.0407 \Omega$$

$$X_{e2} = k^2 X_{e1} = (0.1)^2 \times 4.777 = 0.0478 \Omega$$

4. Summary of Equivalent Circuit Parameters (Referred to Secondary):

- Core Loss Resistance: $R_{c2} = 387.32 \Omega$
- Magnetizing Reactance: $X_{m2} = 233.55 \Omega$
- Equivalent Winding Resistance: $R_{e2} = 0.0407 \Omega$
- Equivalent Leakage Reactance: $X_{e2} = 0.0478 \Omega$

Final Answer: $R_{c2} = 387.32 \Omega$, $X_{m2} = 233.55 \Omega$, $R_{e2} = 0.0407 \Omega$, $X_{e2} = 0.0478 \Omega$.

7.4 2081 Ashwin Exam Paper

Question 7

Explain operation of transformer at loaded condition and prove that main flux in the core of transformer at any loading condition is constant.

Solution:

1. Operation of Transformer Under Load:

1. **No-Load State:** Primary draws no-load current I_0 , establishing main flux ϕ_m and induced back EMF $E_1 \approx V_1$.
2. **Connecting Load:** Secondary load current I_2 flows, producing secondary MMF $N_2 I_2$ and flux ϕ_2 opposing ϕ_m .
3. **Primary Reaction:** Reduction in net core flux lowers primary back EMF E_1 . The resulting voltage difference ($V_1 - E_1$) draws additional primary current I'_2 (load component).
4. **Flux Restoration:** Primary load MMF $N_1 I'_2$ produces flux $\phi'_2 = \phi_2$, which cancels the secondary demagnetizing flux ϕ_2 .

2. Proof of Constant Core Flux ($\phi_m = \text{Constant}$): Applying KVL to the primary winding (neglecting small winding drops):

$$V_1 \approx E_1 = 4.44fN_1\phi_m \implies \phi_m \approx \frac{V_1}{4.44fN_1}$$

Since supply voltage V_1 , frequency f , and primary turns N_1 are constant, main core flux ϕ_m **remains constant** across all load conditions.

Final Answer: Main flux $\phi_m = \frac{V_1}{4.44fN_1}$ is constant because secondary demagnetizing MMF ($N_2 I_2$) is countered by primary load MMF ($N_1 I'_2$).

8 Question 8 (DC Generators, DC Motors & Magnetic Circuits)

8.1 2083 Baishakh Exam Paper

Question 8

A shunt generator supplies 96 A at a terminal voltage of 200 V. The armature and shunt field resistance are $0.1\ \Omega$ and $50\ \Omega$ respectively. The iron and frictional losses are 2500 W. Find:

- a) EMF generated
- b) Commercial efficiency

Solution:

1. Given Parameters:

- Load current $I_L = 96\text{ A}$
- Terminal voltage $V = 200\text{ V}$
- Armature resistance $R_a = 0.1\ \Omega$
- Shunt field resistance $R_{sh} = 50\ \Omega$
- Stray losses (Iron + Friction) = 2500 W

2. Current Calculations:

- Shunt field current:

$$I_{sh} = \frac{V}{R_{sh}} = \frac{200}{50} = 4\text{ A}$$

- Armature current:

$$I_a = I_L + I_{sh} = 96 + 4 = 100\text{ A}$$

3. Generated EMF (E_g):

$$E_g = V + I_a R_a = 200 + (100 \times 0.1) = 200 + 10 = \mathbf{210\text{ V}}$$

4. Power & Loss Calculations:

1. Output Power (P_{out}):

$$P_{\text{out}} = V \cdot I_L = 200 \times 96 = 19,200\text{ W} = 19.2\text{ kW}$$

2. Copper Losses (P_{Cu}):

- Armature copper loss: $I_a^2 R_a = (100)^2 \times 0.1 = 1000\text{ W}$
- Shunt field copper loss: $I_{sh}^2 R_{sh} = (4)^2 \times 50 = 800\text{ W}$

- Total copper loss: $P_{\text{Cu}} = 1000 + 800 = 1800 \text{ W}$

3. **Total Losses (P_{loss}):**

$$P_{\text{loss}} = \text{Stray Losses} + P_{\text{Cu}} = 2500 + 1800 = 4300 \text{ W}$$

4. **Input Power (P_{in}):**

$$P_{\text{in}} = P_{\text{out}} + P_{\text{loss}} = 19,200 + 4300 = 23,500 \text{ W}$$

5. **Commercial Efficiency (η_c):**

$$\eta_c = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\% = \frac{19,200}{23,500} \times 100\% \approx \mathbf{81.70\%}$$

Final Answer: Generated EMF $E_g = \mathbf{210 \text{ V}}$ and Commercial Efficiency $\eta_c = \mathbf{81.70\%}$.

8.2 2082 Bhadra Exam Paper

Question 8

A circular ring of magnetic material has a mean length of 0.5 m, cross sectional area of 100 cm^2 and an air gap of 4 mm. Calculate the MMF required to produce flux of 2 mWb in the air gap, if ring is wound with a coil of 200 turns. Take relative permeability of 800.

Solution:

1. Given Data:

- Mean length of iron ring (l_i) = 0.5 m
- Air gap length (l_g) = 4 mm = 4×10^{-3} m
- Cross-sectional area (A) = $100 \text{ cm}^2 = 0.01 \text{ m}^2$
- Flux (Φ) = 2 mWb = 2×10^{-3} Wb
- Number of turns (N) = 200
- Relative permeability (μ_r) = 800
- Permeability of free space (μ_0) = $4\pi \times 10^{-7} \text{ H/m}$

2. Magnetic Reluctance Calculations:

1. Flux Density (B):

$$B = \frac{\Phi}{A} = \frac{2 \times 10^{-3}}{0.01} = 0.2 \text{ T}$$

2. Iron Core Reluctance ($\mathcal{S}_i$):

$$\mathcal{S}_i = \frac{l_i}{\mu_0 \mu_r A} = \frac{0.5}{(4\pi \times 10^{-7}) \times 800 \times 0.01} \approx 49,735.9 \text{ A} \cdot \text{t/Wb}$$

3. Air Gap Reluctance ($\mathcal{S}_g$):

$$\mathcal{S}_g = \frac{l_g}{\mu_0 A} = \frac{4 \times 10^{-3}}{(4\pi \times 10^{-7}) \times 0.01} \approx 318,309.9 \text{ A} \cdot \text{t/Wb}$$

4. Total Magnetic Reluctance ($\mathcal{S}_{\text{total}}$):

$$\mathcal{S}_{\text{total}} = \mathcal{S}_i + \mathcal{S}_g = 49,735.9 + 318,309.9 = 368,045.8 \text{ A} \cdot \text{t/Wb}$$

3. Total Required MMF:

$$\text{MMF} = \Phi \times \mathcal{S}_{\text{total}} = (2 \times 10^{-3}) \times 368,045.8 \approx \mathbf{736.09 \text{ A} \cdot \text{t}}$$

Final Answer: Required MMF = **736.09 A · t**.

8.3 2082 Baishakh Exam Paper

Question 8

Explain the production of torque in DC motor with necessary mathematical explanation and derivation. What is the role of back emf in DC motor?

Solution:

Part A: Derivation of Electromagnetic Torque Equation

1. **Physical Principle:** A current-carrying conductor in a magnetic field experiences Lorentz force $F = B I_c l$.
2. **Mathematical Derivation:** Let P = poles, Φ = flux/pole, Z = total conductors, A = parallel paths, I_a = armature current, r = armature radius, l = active length.

- Conductor current $I_c = \frac{I_a}{A}$ and Flux density $B = \frac{P\Phi}{2\pi r l}$.
- Torque on a single conductor:

$$T_c = F_c \times r = (B I_c l) \cdot r = \left(\frac{P\Phi}{2\pi r l} \right) \left(\frac{I_a}{A} \right) l r = \frac{P\Phi I_a}{2\pi A}$$

- Total electromagnetic torque T_a for all Z conductors:

$$T_a = Z \cdot T_c = \frac{P\Phi Z I_a}{2\pi A} = 0.159 \left(\frac{PZ}{A} \right) \Phi I_a \quad (\text{N} \cdot \text{m})$$

Part B: Role of Back EMF (E_b)

1. **Self-Regulating Governor:** Armature current is governed by $I_a = \frac{V - E_b}{R_a}$.
 - Load increases $\rightarrow$ speed N drops $\rightarrow E_b$ drops $\rightarrow I_a$ increases to develop necessary torque.
 - Load decreases $\rightarrow$ speed N rises $\rightarrow E_b$ increases $\rightarrow I_a$ decreases.
2. **Energy Conversion Mechanism:** Multiplying voltage equation by I_a :

$$V I_a = E_b I_a + I_a^2 R_a$$

where $V I_a$ is electrical power input, $I_a^2 R_a$ is copper loss, and $E_b I_a$ represents converted mechanical power.

Final Answer: Electromagnetic Torque: $\mathbf{T_a = \frac{P\Phi Z I_a}{2\pi A} = 0.159 \left(\frac{PZ}{A} \right) \Phi I_a \text{ N} \cdot \text{m}}$. Back EMF acts as a self-regulating governor and mechanical power converter ($P_{\text{mech}} = E_b I_a$).

8.4 2081 Ashwin Exam Paper

Question 8

A 240 V DC shunt motor has armature winding resistance of $0.4\ \Omega$ and field winding resistance of $120\ \Omega$. It draws a current of 30 A at half load and the corresponding speed is 1400 rpm. If a resistance of $1.2\ \Omega$ is connected in series with the armature winding and load torque is decreased by 20%, calculate the new speed.

Solution:

1. Given Data:

- Supply voltage $V = 240\text{ V}$, $R_a = 0.4\ \Omega$, $R_{sh} = 120\ \Omega$
- Initial state (Half Load): $I_{L1} = 30\text{ A}$, $N_1 = 1400\text{ rpm}$
- Modified state: $R_{\text{ext}} = 1.2\ \Omega$ in series with armature, $T_2 = 0.8T_1$

2. Initial State Calculations (Condition 1):

- Shunt field current:

$$I_{sh} = \frac{V}{R_{sh}} = \frac{240}{120} = 2\text{ A} \quad (\text{constant field flux } \Phi)$$

- Initial armature current:

$$I_{a1} = I_{L1} - I_{sh} = 30 - 2 = 28\text{ A}$$

- Initial Back EMF (E_{b1}):

$$E_{b1} = V - I_{a1}R_a = 240 - (28 \times 0.4) = 240 - 11.2 = 228.8\text{ V}$$

3. Modified State Calculations (Condition 2):

- New Armature Current (I_{a2}):** Since Φ is constant, torque $T \propto I_a$:

$$\frac{T_2}{T_1} = \frac{I_{a2}}{I_{a1}} \implies 0.8 = \frac{I_{a2}}{28} \implies I_{a2} = 0.8 \times 28 = \mathbf{22.4\text{ A}}$$

- Total Armature Resistance (R_{a2}):**

$$R_{a2} = R_a + R_{\text{ext}} = 0.4 + 1.2 = 1.6\ \Omega$$

- New Back EMF (E_{b2}):**

$$E_{b2} = V - I_{a2}R_{a2} = 240 - (22.4 \times 1.6) = 240 - 35.84 = \mathbf{204.16\text{ V}}$$

- Calculating New Speed (N_2):** Since $E_b \propto N$ (for constant flux Φ):

$$\frac{E_{b2}}{E_{b1}} = \frac{N_2}{N_1} \implies \frac{204.16}{228.8} = \frac{N_2}{1400}$$

$$N_2 = 1400 \times \left(\frac{204.16}{228.8} \right) \approx \mathbf{1249.25\text{ rpm}}$$

Final Answer: New Motor Speed $N_2 = \mathbf{1249.25\text{ rpm}}$.

9 Question 9 (Induction Motors & Transformer Efficiency)

9.1 2083 Baishakh Exam Paper

Question 9

A three-phase induction motor is wound for 4 poles and supplied from 50 Hz supply system. Calculate:

- a) The speed of motor when the slip is 5%
- b) The rotor emf frequency when the motor runs at 600 rpm.

Solution:

1. Given Parameters & Synchronous Speed:

- Number of poles $P = 4$, Supply frequency $f = 50$ Hz
- Synchronous speed (N_s):

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500 \text{ rpm}$$

2. Part (a): Motor Speed (N) at Slip $s = 5\% = 0.05$:

$$N = N_s(1 - s) = 1500 \times (1 - 0.05) = 1500 \times 0.95 = \mathbf{1425 \text{ rpm}}$$

3. Part (b): Rotor EMF Frequency (f') at $N = 600$ rpm:

1. Calculate slip (s):

$$s = \frac{N_s - N}{N_s} = \frac{1500 - 600}{1500} = \frac{900}{1500} = 0.6 \quad (60\%)$$

2. Rotor frequency (f'):

$$f' = s \cdot f = 0.6 \times 50 = \mathbf{30 \text{ Hz}}$$

Final Answer: a) Motor Speed $N = \mathbf{1425 \text{ rpm}}$ b) Rotor EMF Frequency $f' = \mathbf{30 \text{ Hz}}$.

9.2 2082 Bhadra Exam Paper

Question 9

A 150 kVA single phase transformer has an iron loss of 700 W and a full load copper loss of 1800 W. Calculate the copper losses, iron losses, output power and efficiency of transformer at 0.8 power factor lagging when secondary is 25% overloaded.

Solution:

1. Given Data:

- Full-load rating $S = 150 \text{ kVA}$, Power factor $\cos \phi = 0.8$ lagging
- Iron loss $P_i = 700 \text{ W} = 0.7 \text{ kW}$
- Full-load copper loss $P_{cu,FL} = 1800 \text{ W} = 1.8 \text{ kW}$
- Loading factor for 25% overload: $x = 1 + 0.25 = 1.25$

2. Losses and Power Calculations at 25% Overload ($x = 1.25$):

1. **Iron Loss (P_i):** Constant loss independent of loading:

$$P_i = 700 \text{ W} = 0.7 \text{ kW}$$

2. **Copper Loss ($P_{cu,x}$):** Copper loss varies with the square of load fraction x :

$$P_{cu,x} = x^2 \times P_{cu,FL} = (1.25)^2 \times 1800 = 1.5625 \times 1800 = 2812.5 \text{ W} = 2.8125 \text{ kW}$$

3. **Output Power (P_{out}):**

$$P_{out} = x \times S \times \cos \phi = 1.25 \times 150 \times 0.8 = 150 \text{ kW}$$

4. **Total Losses (P_{total}):**

$$P_{total} = P_i + P_{cu,x} = 0.7 + 2.8125 = 3.5125 \text{ kW}$$

3. Efficiency Calculation (η):

$$\eta = \left(\frac{P_{out}}{P_{out} + P_{total}} \right) \times 100\% = \left(\frac{150}{150 + 3.5125} \right) \times 100\% = \frac{150}{153.5125} \times 100\% \approx 97.71\%$$

Final Answer: Iron Loss $P_i = 700 \text{ W}$, Copper Loss $P_{cu,x} = 2812.5 \text{ W}$, Output Power $P_{out} = 150 \text{ kW}$, Efficiency $\eta = 97.71\%$.

9.3 2082 Baishakh Exam Paper

Question 9

How does three phase induction motor start? Why speed of this motor never reaches synchronous speed?

Solution:

1. Starting Mechanism of a Three-Phase Induction Motor:

1. **Creation of Rotating Magnetic Field (RMF):** When balanced 3-phase AC currents flow through stator windings, a magnetic field of constant magnitude rotating at synchronous speed $N_s = \frac{120f}{P}$ is produced in the air gap.
2. **Electromotive Force (EMF) Induction in Rotor:** The RMF cuts the stationary rotor conductors, inducing an EMF per Faraday's law of electromagnetic induction.
3. **Rotor Current & Torque Generation:** Since the rotor circuit forms a closed path (via end rings or external resistance), the induced EMF drives short-circuit rotor currents. These currents interact with the stator magnetic field, creating an electromagnetic force ($F = BIL$) and driving torque that accelerates the rotor in the direction of the RMF (Lenz's Law).

2. Why Motor Speed Never Reaches Synchronous Speed ($N < N_s$):

1. **Principle of Relative Speed:** Induction motor action strictly depends on the relative motion ($N_s - N$) between stator RMF and rotor conductors.
2. **If Rotor Attained Synchronous Speed ($N = N_s$):**
 - Relative speed ($N_s - N$) = 0.
 - No magnetic flux lines cut the rotor conductors.
 - Induced EMF in rotor drops to zero ($e = 0$).
 - Rotor current becomes zero ($i = 0$), causing electromagnetic torque to drop to zero ($T = 0$).
3. **Conclusion:** Without driving torque, mechanical friction and windage immediately slow the rotor down below N_s . Hence, an induction motor **must always operate at a speed less than synchronous speed** ($N < N_s$) to maintain non-zero slip ($s > 0$).

Final Answer: Starting: Stator RMF (N_s) $\rightarrow$ Rotor flux cutting $\rightarrow$ Induced EMF/Current $\rightarrow$ Driving Torque ($F = BIL$). Motor cannot reach N_s because at $N = N_s$, relative speed = 0 $\implies e = 0, T = 0$.

9.4 2081 Ashwin Exam Paper

Question 9

Explain the Torque-speed characteristics of three phase induction motor with the help of proper mathematics and graph.

Solution:

1. Mathematical Derivation of Electromagnetic Torque Equation: The electromagnetic torque T developed by a 3-phase induction motor per phase is:

$$T = \frac{k s E_2^2 R_2}{R_2^2 + (sX_2)^2}$$

where $s = \frac{N_s - N}{N_s}$ is slip, E_2 is standstill rotor EMF per phase, R_2 is rotor resistance per phase, X_2 is standstill rotor reactance per phase, and $k = \frac{3}{2\pi N_s / 60}$ is constant.

2. Analysis of Torque-Speed / Torque-Slip Operating Regions:

1. **Low-Slip Region** ($s \approx 0$, near N_s): Since s is very small, $(sX_2)^2 \ll R_2^2$:

$$T \approx \left(\frac{kE_2^2}{R_2} \right) s \implies T \propto s$$

The torque is directly proportional to slip s . The characteristic is a **straight line** (Stable Region).

2. **High-Slip Region** ($s \approx 1$, near standstill $N = 0$): Since s is large, $(sX_2)^2 \gg R_2^2$:

$$T \approx \frac{k s E_2^2 R_2}{(sX_2)^2} = \left(\frac{kE_2^2 R_2}{X_2^2} \right) \frac{1}{s} \implies T \propto \frac{1}{s}$$

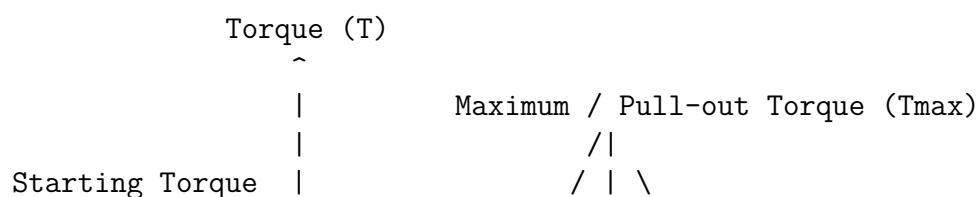
The torque is inversely proportional to slip s . The characteristic forms a **rectangular hyperbola** (Unstable Region).

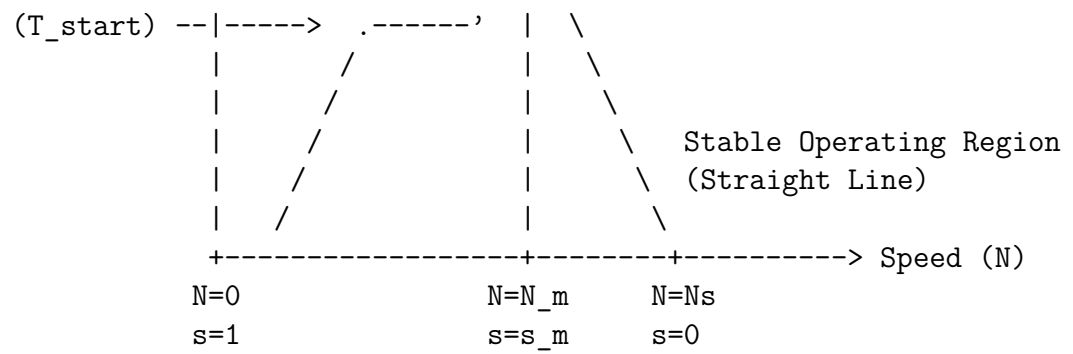
3. **Maximum / Breakdown Torque** ($T_{\max}$): Differentiating T with respect to s gives maximum torque condition at slip:

$$s_m = \frac{R_2}{X_2} \implies T_{\max} = \frac{kE_2^2}{2X_2}$$

Maximum torque is independent of rotor resistance R_2 , though R_2 shifts the slip s_m at which breakdown torque occurs.

3. Torque-Speed / Torque-Slip Characteristic Curve:





- **Stable Region ($s < s_m$):** Motor automatically adjusts speed to match load torque changes.
- **Unstable Region ($s > s_m$):** If load torque exceeds T_{max} , the motor stalls.

Final Answer: Torque Equation: $T = \frac{ksE_2^2R_2}{R_2^2 + (sX_2)^2}$. Low slip: $T \propto s$ (linear). High slip: $T \propto 1/s$ (hyperbola). Max torque at $s_m = R_2/X_2$.

10 Question 10 (Short Notes & Conceptual Explanations)

10.1 2083 Baishakh Exam Paper

Question 10

Briefly explain (Any Three):

- a) Series and parallel magnetic circuit
- b) Universal motor
- c) DC motor starter
- d) Voltage regulation of transformer

Solution:

a) Series and Parallel Magnetic Circuit:

- **Series Magnetic Circuit:** A magnetic circuit where the same flux Φ flows through all series-connected sections (e.g., iron core and air gap). Total reluctance is the sum of individual reluctances:

$$\mathcal{S}_{\text{total}} = \mathcal{S}_1 + \mathcal{S}_2 + \cdots + \mathcal{S}_n$$

Total required MMF: $\text{MMF}_{\text{total}} = \Phi \mathcal{S}_{\text{total}} = H_1 l_1 + H_2 l_2 + \cdots + H_n l_n$.

- **Parallel Magnetic Circuit:** A magnetic circuit providing two or more independent paths for magnetic flux. Total flux divides among parallel branches:

$$\Phi = \Phi_1 + \Phi_2$$

MMF across parallel branches remains identical: $\text{MMF} = \Phi_1 \mathcal{S}_1 = \Phi_2 \mathcal{S}_2$.

b) Universal Motor:

- **Definition:** A specialized motor designed to operate efficiently on both single-phase AC supply and DC supply.
- **Construction & Working:** It is essentially a modified DC series motor with laminated field cores and armature to minimize eddy current losses. Under AC excitation, both field flux Φ and armature current I_a reverse direction simultaneously every half-cycle, maintaining unidirectional torque ($T \propto \Phi I_a > 0$).
- **Applications:** Portable hand drills, vacuum cleaners, mixers/grinders, sewing machines.

c) DC Motor Starter:

- **Need for Starter:** At standstill ($N = 0$), back EMF $E_b = 0$. Starting armature current $I_a = \frac{V}{R_a}$ is dangerously high (5 to 20 times rated current) because armature resistance R_a is extremely small.
- **Function:** A starter connects variable resistance in series with the armature to limit starting current. As the motor accelerates and builds up back EMF E_b , the external resistance is progressively cut out until normal running speed is achieved.

d) Voltage Regulation of Transformer:

- **Definition:** The change in secondary terminal voltage from no-load (V_{20}) to full-load (V_2) expressed as a fraction or percentage of full-load voltage, maintaining constant primary supply voltage.
- **Formula:**

$$\% \text{ Regulation} = \frac{V_{20} - V_2}{V_2} \times 100\% \approx \frac{I_2 R_{e2} \cos \phi \pm I_2 X_{e2} \sin \phi}{V_2} \times 100\%$$

where $+$ is used for lagging power factor and $-$ for leading power factor.

Final Answer: Covered: Series/Parallel magnetic circuits (Φ vs MMF division), Universal motor (AC/DC series motor), DC starter ($I_{a,\text{start}} = V/R_a$), Voltage Regulation ($\frac{V_{20}-V_2}{V_2} \times 100\%$).

10.2 2082 Bhadra Exam Paper

Question 10

Justify and explain (Any Three):

- a) Speed of three phase induction motor is always less than synchronous speed
- b) Iron losses of transformer is independent of load
- c) DC motor draws high starting current when switched on directly
- d) Single phase induction motor has two stator winding

Solution:

a) Speed of 3- ϕ Induction Motor is Always Less Than Synchronous Speed ($N < N_s$):

- **Justification:** Induction motor torque relies strictly on Faraday's law of electromagnetic induction.
- **Reasoning:** Induced EMF/current in the rotor requires relative motion ($N_s - N$) between stator RMF (N_s) and rotor (N). If $N = N_s$, relative speed becomes zero $\implies$ flux cutting = 0 $\implies$ induced EMF $e = 0 \implies$ torque $T = 0$. Friction slows the rotor down, ensuring $N < N_s$ always.

b) Iron Losses of Transformer is Independent of Load:

- **Justification:** Iron/core losses ($P_h \propto B_m^{1.6} f$ and $P_e \propto B_m^2 f^2$) depend only on peak core flux density B_m and supply frequency f .
- **Reasoning:** Since primary supply voltage $V_1 \approx 4.44 f N_1 \Phi_m$ is constant, core flux Φ_m remains constant from no-load to full-load. Hence, iron losses remain constant regardless of loading.

c) DC Motor Draws High Starting Current When Switched On Directly:

- **Justification:** Armature current is governed by $I_a = \frac{V - E_b}{R_a}$.
- **Reasoning:** At start ($t = 0$), rotor speed $N = 0 \implies$ back EMF $E_b = 0$. Since armature resistance R_a is intentionally very small ($< 1 \Omega$), direct online switching causes $I_{a,\text{start}} = \frac{V}{R_a}$, drawing 5 to 20 times rated full-load current.

d) Single-Phase Induction Motor Has Two Stator Windings:

- **Justification:** A single-phase winding creates a pulsating field with zero net starting torque.
- **Reasoning:** According to double-field revolving theory, a pulsating field decomposes into two equal fields rotating in opposite directions, producing equal and opposite torques at standstill ($T_{\text{start}} = 0$). An auxiliary starting winding displaced by 90° electrical creates a rotating magnetic field at startup to make the motor self-starting.

Final Answer: Justified: $N < N_s$ for relative flux cutting; Iron loss constant because $\Phi_m \propto V_1/f$ is constant; $I_{a,\text{start}} = V/R_a$ high because $E_b = 0$; Auxiliary winding creates starting RMF.

10.3 2082 Baishakh Exam Paper

Question 10

Write notes on: (Any Two)

- a) Magnetization of magnetic material
- b) Auto transformer and its application
- c) Stepper motor

Solution:

a) Magnetization of Magnetic Material:

- **Concept:** Applying magnetic field intensity H aligns magnetic domains in ferromagnetic materials, establishing flux density B .
- **B-H Curve:** Exhibits a linear region ($\mu = \text{const}$), knee point (incipient saturation), and saturation region (where B levels off as all domains align).
- **Hysteresis:** Lagging of flux density B behind field intensity H during AC magnetization cycles, causing core power loss.

b) Auto Transformer and Its Applications:

- **Definition:** A transformer with a single continuous winding tapped at intermediate points, acting as both primary and secondary.
- **Working:** Power is transferred partly by electrical conduction through common turns and partly by electromagnetic induction.
- **Advantages:** Requires less copper, lower cost, smaller footprint, higher efficiency, and better voltage regulation than 2-winding transformers.
- **Applications:** Variacs (variable AC supply in labs), AC motor starting compensators, grid interconnectors.

c) Stepper Motor:

- **Definition:** An electromechanical actuator converting discrete digital electrical pulses into precise angular shaft steps.
- **Types:** Permanent Magnet (PM), Variable Reluctance (VR), and Hybrid Stepper Motors.
- **Key Features:** Precise open-loop position control without feedback sensors.
- **Applications:** 3D printers, CNC machines, robotics, disk drives, camera lenses.

Final Answer: Magnetization (B-H curve & domain alignment); Auto Transformer (single winding, conductive + inductive transfer); Stepper Motor (digital pulse to step angle conversion).

10.4 2081 Ashwin Exam Paper

Question 10

Write short notes on (Any Two):

- a) Role of commutator and carbon brush in DC machine
- b) Hysteresis and eddy current power losses
- c) Stepper motor

Solution:

a) Role of Commutator and Carbon Brush in DC Machine:

- **Commutator:** Mechanical rectifier in DC generators (converts AC armature voltage to DC) and torque inverter in DC motors (reverses coil current to maintain continuous unidirectional torque).
- **Carbon Brushes:** Sliding contacts collecting current from (or feeding current to) commutator segments; carbon/graphite provides self-lubricating smooth contact with low wear.

b) Hysteresis and Eddy Current Power Losses:

- **Hysteresis Loss (P_h):** Power lost in overcoming magnetic domain friction during cyclic magnetization. $P_h = \eta B_m^{1.6} f V W$. Minimized using high-grade silicon steel core.
- **Eddy Current Loss (P_e):** Power lost due to induced circulating currents in the conductive metallic core. $P_e = K_e B_m^2 f^2 t^2 V W$. Minimized by laminating the iron core into thin, insulated sheets.

c) Stepper Motor:

- Converts input digital pulses into precise discrete mechanical step angles ($0.9^\circ, 1.8^\circ, 7.5^\circ$).
- Operates reliably in open-loop control systems without position feedback encoders. Used widely in robotics, 3D printing, and automation.

Final Answer: Commutator (AC/DC conversion); Carbon Brushes (sliding current collector); Hysteresis loss ($P_h \propto B_m^{1.6} f$) & Eddy loss ($P_e \propto B_m^2 f^2 t^2$); Stepper motor (digital step angle positioner).